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Dependent by Design

US Tech Integration and the Erosion of Digital Sovereignty in Nordic EU Startups

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Abstract

This thesis investigates the relationship between United States tech dependencies and data collection practices in Nordic startups operating under EU/EEA data protection law, with a focus on Digital Sovereignty Theory. Motivated by the tension between the scalability benefits of US-based services and the digital sovereignty concerns of the EU, this study addresses a critical research gap: the lack of empirical validation of Digital Sovereignty Theory predictions in the Nordic startup context. The study examines how reliance on US-based third-party services (*e.g.*, Amazon Web Services, Google Cloud, Meta) conflicts with General Data Protection Regulation compliance and EU digital sovereignty initiatives like Gaia-X. Using a weighted scoring system for data collection practices and third-party service dependencies, the research examines the data governance consequences of Nordic startups' reliance on US infrastructure. Analysis covers 501 Nordic startups with iOS apps, using Privacy Nutrition Labels and privacy policy data collected May 2-4, 2026. Findings highlight a statistically significant positive relationship ($\beta = 0.571$, $p < 0.001$) between US service dependency and data sensitivity scores, driven primarily by Data and AI services, with Nordic startups using 6.8× more US services than EU services (75.2% use no EU services at all). Notably, Sensitive Personal Data collection (GDPR Article 9) shows no significant association with US dependency ($p = 0.477$), a pattern suggesting startups expand collection of less-regulated categories (General Personal and Usage Data) while strategically avoiding the Article 9 categories that carry the highest legal risk. The model explains 18% of variance (R^2=0.180), with business model (Marketplace, B2C) and firm size as comparably strong predictors alongside US dependency. The findings inform policymakers and industry stakeholders navigating the balance between innovation and data protection.

Keywords: Digital Sovereignty, GDPR, US Tech Dependencies, Nordic Startups, Data Collection, Privacy Nutrition Labels

Sammanfattning

Denna avhandling undersöker sambandet mellan beroendet av amerikansk teknik och datainsamlingspraktiker bland nordiska EU-startups, med särskilt fokus på Digital Suveränitetsteori. Motiverad av spänningen mellan skalbarhetsfördelarna med amerikansk teknik och EU:s oro för digital suveränitet, adresserar denna studie ett kritiskt forskningsgap: bristen på empirisk testning av Digital Suveränitetsteoris prediktioner i den nordiska startup-kontexten. Studien granskar hur beroendet av amerikanska tjänster (*e.g.* Amazon Web Services, Google Cloud, Meta) konflikerar med GDPR-efterlevnad och EU:s initiativ för digital suveränitet som Gaia-X. Genom att använda ett viktat poängsystem för datainsamling och beroende av tredjepartstjänster, undersöker forskningen datastyrningskonsekvenserna av nordiska startups beroende av amerikansk infrastruktur. Analysen omfattar 501 nordiska startups med iOS-appar, med hjälp av Privacy Nutrition Labels och integritetspolicydata insamlad 2–4 maj 2026. Resultaten visar på ett statistiskt signifikant positivt samband ($\beta = 0{,}571$, $p < 0{,}001$) mellan beroende av amerikanska tjänster och känslighetspäng för datainsamling, drivet främst av Data- och AI-tjänster, där nordiska startups använder 6,8× fler amerikanska tjänster än EU-tjänster (75,2% använder inga EU-tjänster alls). Anmärkningsvärt är att insamling av Känsliga Personuppgifter (GDPR Artikel 9) inte visar något signifikant samband med amerikanskt beroende ($p = 0{,}477$), vilket tyder på strategisk efterlevnadsmedvetenhet: startups ökar insamlingen av Allmänna Personuppgifter och Användningsdata men undviker juridiskt reglerade kategorier. Modellen förklarar 18% av variansen ($R^2 = 0{,}180$). Forskningen erbjuder insikter till beslutsfattare och branschaktörer om att balansera innovation med dataskydd.

Nyckelord: Digital Suveränitet, GDPR, Amerikanska Teknikberoenden, Nordiska Startups, Datainsamling, Privacy Nutrition Labels

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Abbreviations

AI Artificial Intelligence
AWS Amazon Web Services
B2C Business to Consumer
B2G Business to Government
CDN Content Delivery Network
CLOUD Act Clarifying Lawful Overseas Use of Data Act
EEA European Economic Area
IT Information Technology
NSA National Security Agency
RBV Resource-Based View
TCE Transaction Cost Economics
ANOVA Analysis of Variance
OLS Ordinary Least Squares
EDPB European Data Protection Board
EU European Union
GDPR General Data Protection Regulation
IP Internet Protocol
IRB Institutional Review Board
OS Operating System
PNL Privacy Nutrition Label
SaaS Software as a Service
SME Small and Medium-sized Enterprise
US United States
VIF Variance Inflation Factor
FISA Foreign Intelligence Surveillance Act
IMY Integritetsmyndigheten
JSON JavaScript Object Notation
LLM Large Language Model
NLP Natural Language Processing
OECD Organisation for Economic Co-operation and Development
SCC Standard Contractual Clauses

SEC Securities and Exchange Commission

LOESS Locally Estimated Scatterplot Smoothing

IRR Incidence Rate Ratio

IDs Identifiers

CTO Chief Technology Officer

DPA Data Processing Agreement

DST Digital Sovereignty Theory

API Application Programming Interface

HC1 Heteroskedasticity-Consistent standard errors (type 1)

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Foreword

Technology as well as science are neutral by definition. Therefore, it is my belief that it is every entrepreneur's and every scientist's responsibility to deploy these tools for the benefit of the people. Standing on the shoulders of giants, technology has become magic and science has become fortune-telling. We worship them blindly, the sorcerers of Oz. God is dead but one more time demi-gods walk amongst us, yet we stand to discover again that virtue is the fundament of civilization.

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Lukas Moers
Stockholm, Sweden
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Chapter 1

Introduction

1.1 Background

The digital transformation of Nordic startups represents a paradox of modern technological development. On one hand, the region has emerged as a global innovation leader, producing unicorns across fintech (Klarna, iZettle), health tech (Kry, Oura), Software as a Service (SaaS) (Mentimeter, Spotify), and mobility and e-commerce (Voi, Boozt) (Autio & Rannikko, 2016; Organisation for Economic Co-Operation and Development, 2024). Nordic startups have raised billions in venture capital, with Sweden, Finland, Denmark, and Norway consistently ranking among Europe's most innovative ecosystems (Nordic Council of Ministers, 2023). This success stems from a unique combination of strong digital infrastructure, high English proficiency, supportive policy environments, and a culture of collaboration between public and private sectors.

On the other hand, this rapid digitalization has come with a hidden cost: increasing reliance on US-based third-party services for essential digital infrastructure. US technology providers such as Amazon Web Services, Google Cloud, Microsoft Azure, Firebase, Google Analytics, and Meta's advertising platforms have become the de facto standard for Nordic startup technology stacks (Bradford, 2023; Glasze et al., 2023). These services offer undeniable advantages in scalability, cost-efficiency, feature richness, and global reach, critical factors for startups competing in international markets. However, as Zuboff (2022) argues in her analysis of surveillance capitalism, this reliance creates structural dependencies that may undermine the very autonomy and innovation capacity that define the Nordic digital success story.

The tension between these benefits and risks forms the core of this thesis. While US services enable Nordic startups to compete globally and access cutting-edge technologies, they also raise serious concerns about data sovereignty, General Data Protection Regulation (GDPR) compliance, and the long-term resilience of the European Union's digital infrastructure. As Glasze et al. (2023) demonstrate in their analysis of contested digital spatialities, digital sovereignty is not merely a technical issue but a geopolitical one that reflects broader power dynamics in the global digital economy. For Nordic startups, this tension represents a fundamental challenge to their ability to maintain control over their digital destiny while participating in the global digital ecosystem.

Three concepts recur throughout this thesis and are defined here at the outset. A *startup* is understood as a young, growth-oriented firm developing a technology-based product or service; operationally, this study identifies startups through thehub.io registration, with iOS app presence serving as the measurement criterion required to access standardized Privacy Nutrition Label disclosures at scale (see Chapter 4). *Third-party services* are externally provided software components (such as cloud hosting, analytics, payment, and advertising services) that

a startup integrates into its product rather than building in-house. *Digital sovereignty*, following Pohle and Thiel (2020), denotes the capacity of an actor (a firm, state, or union) to exercise autonomy, control, and resilience over its digital infrastructure and data.

1.2 Research Gap and Problem Statement

Nordic startups face a fundamental challenge in balancing the need for scalable, cost-effective US technology services with the requirements of EU digital sovereignty and GDPR compliance. This tension manifests in multiple ways. First, jurisdictional conflicts arise from the Clarifying Lawful Overseas Use of Data Act (CLOUD Act), which enables US law enforcement to compel disclosure of data from any provider under US jurisdiction (including EU subsidiaries of US-controlled entities) regardless of where the data is physically stored (Christakis, 2019; European Data Protection Board & European Data Protection Supervisor, 2019). While the Act's application to truly independent EU-based entities remains legally contested (Rojszczak, 2020), the European Data Protection Board (EDPB) has concluded that US-controlled providers cannot guarantee GDPR compliance when subject to CLOUD Act compulsion orders (European Data Protection Board & European Data Protection Supervisor, 2019). Second, the market dominance of US providers in cloud infrastructure (e.g., Amazon Web Services (AWS), Google Cloud, Azure), analytics (e.g., Google Analytics, Mixpanel), and AI/ML (e.g., TensorFlow, PyTorch) creates structural dependencies that are difficult for startups to avoid without sacrificing competitiveness. Third, path dependence and vendor lock-in trap startups once they commit to a US-based service. Switching requires data migration, staff retraining, and system rearchitecture, costs that are prohibitively high (David, 1985; Eisenmann et al., 2009). Finally, GDPR compliance becomes more complex when using US services, as startups must navigate both EU data protection requirements and potential US legal requests, creating a complex and often contradictory compliance landscape.

The central problem is that while US services provide the scalability and functionality that Nordic startups need to compete globally, they also create dependencies that may undermine the very autonomy, control, and resilience that the EU seeks to promote through its digital sovereignty initiatives.

The theoretical foundations for understanding this tension are well-established. Digital Sovereignty Theory provides a comprehensive framework for analyzing autonomy, control, and resilience in digital ecosystems (Pohle & Thiel, 2020), operationalized for service-layer analysis by Fries et al. (2023). Bradford (2023) extends this analysis to the geopolitical level, demonstrating how the "Brussels Effect" of EU regulation shapes global digital governance while simultaneously creating tensions with US technological dominance. Zuboff (2022) critique of surveillance capitalism further illuminates the power asymmetries inherent in the relationship between European startups and US technology platforms, where data collection practices become embedded in the very infrastructure of digital innovation.

Despite these robust theoretical frameworks, empirical validation in the Nordic startup context remains limited. Prior research has primarily taken three forms: qualitative case analyses that provide rich contextual insights but offer limited generalizability (Garcia Caceres, 2025; Roberts, 2021); theoretical discussions that lack empirical validation of predictions (DeNardis, 2014; Pohle & Thiel, 2020); and policy-focused analyses that examine regulatory frameworks without systematically assessing their impact on actual business practices (European Commission, 2022; European Data Protection Board, 2021).

Several critical gaps persist in the literature. First, few studies have examined the actual relationship between service dependencies and data collection practices at scale using systematic quantitative methods. Second, while GDPR establishes a clear hierarchy of data sensitivity,

existing research has not systematically operationalized this framework into a weighted scoring system for comparative analysis. This thesis addresses this limitation by developing a GDPR-weighted data sensitivity score that reflects the legal risk hierarchy established in Articles 4 and 9 of the GDPR. Third, most empirical studies on digital sovereignty have focused on larger EU member states (Germany, France) or the EU as a whole, leaving the Nordic context (with its distinctive combination of advanced digital economies, strong privacy cultures, and small domestic markets) understudied. Finally, existing research tends to treat third-party services as a homogeneous category, without distinguishing between different types of services (infrastructure, data and AI, marketing) that may have varying implications for digital sovereignty. The analysis addresses this gap by decomposing the relationship by service type using Fries's layer model.

This thesis addresses these gaps through a novel methodology that integrates three key innovations. First, it develops weighted scoring systems grounded in GDPR's legal framework and Fries's layer model, enabling nuanced measurement of both data sensitivity and service dependency risks. Second, it employs an innovative multi-agent validation approach that leverages multiple language models to achieve consensus-based validation of third-party service identification. Third, the large-scale empirical analysis, the first quantitative study of its kind in the Nordic context, examines 501 Nordic startups (the sample that remained after the multi-step construction and validation pipeline detailed in Chapter 4) to provide generalizable insights into the relationship between US service dependencies and data collection practices. The contribution is thus primarily *empirical*, the first large-scale quantitative test of Digital Sovereignty Theory predictions in the Nordic startup context, with a *methodological* component in the GDPR-weighted scoring systems and the multi-agent validation pipeline.

The broader implications of this research extend beyond the Nordic context. As Bradford (2023) argues, the tension between US technological dominance and EU regulatory ambition represents a fundamental fault line in the global digital economy. The Nordic startup ecosystem, with its combination of technological sophistication and regulatory alignment with the EU, serves as a critical case study for understanding how this tension plays out in practice. The findings inform ongoing policy debates about digital sovereignty, data protection, and the future of the transatlantic digital relationship.

Framed within industrial management, this thesis speaks to how technology-based firms govern their sourcing and supplier relationships. The choice of third-party digital infrastructure is fundamentally a strategic make-or-buy and dependency-management decision, with consequences for firm autonomy, regulatory risk exposure, and long-term operational resilience. By quantifying how service-sourcing decisions relate to data-collection practices, the study informs technology and operations strategy for startups, and for the managers, advisors, and investors who shape those decisions, thereby linking the digital-sovereignty debate to core questions of industrial management.

1.3 Purpose and Aim

The aim of this study is to empirically test Digital Sovereignty Theory predictions regarding the tension between US technology dependencies and data governance in Nordic startups operating under EU/European Economic Area (EEA) data protection law. Specifically, this study seeks to quantify the relationship between US third-party service dependency and GDPR-weighted data sensitivity scores, identify which categories of third-party services drive this relationship most strongly, assess how this relationship varies by startup characteristics such as industry, size, age, funding, and business model, and discuss implications for startups navigating digital sovereignty challenges.

1.4 Research Questions (RQ)

This study addresses the following primary research question:

> *How do US third-party services relate to Nordic startups' data collection practices?*

To operationalize this question, the study examines three specific dimensions: (RQ1) Do Nordic startups with higher US service dependency scores collect more sensitive data, as measured by GDPR-weighted scores? (RQ2) Which categories of third-party services, Infrastructure, Data and AI, or Marketing, drive this relationship most strongly? (RQ3) Does this relationship vary by startup characteristics such as industry, size, age, funding, and business model?

1.5 Delimitations

This study is subject to several scope limitations. Geographically, the analysis covers only the five Nordic countries (Denmark, Finland, Iceland, Norway, Sweden). Platform-wise, only iOS applications are included, as Android and web data were unavailable at the scale required. The startup population is defined by thehub.io registration (which requires self-identification as a startup or growth-stage venture) combined with the presence of an iOS application as a proxy for active product development. This operationalization was chosen because it enables large-scale, programmatic data collection of standardised privacy disclosures (PNLs) that are not available for web-only or pre-launch firms. The iOS-app criterion may introduce modest selection bias toward consumer-facing and more product-mature startups, since earlier-stage B2B ventures may not yet have a mobile application; this directional bias is noted as a limitation in Chapter 7. Importantly, large established companies with iOS apps are largely excluded by the thehub.io registration requirement, which functions as the primary startup qualifier. The study deliberately spans multiple industries (for example SaaS, fintech, and health tech) rather than focusing on a single sector: digital-sovereignty pressures and US service dependency are economy-wide phenomena, and a cross-industry sample is necessary both to assess the generality of the relationship and to obtain sufficient statistical power. Industry is retained as a control variable throughout the analysis, so that sector-specific effects can be separated from the overall relationship. Data collection practices are measured exclusively through Privacy Nutrition Labels, which are self-reported and may underrepresent actual collection. The analysis is cross-sectional, drawing on data collected 2–4 May 2026, and third-party service identification relies on disclosures in startup privacy policies.

1.6 Sustainability and the UN Sustainable Development Goals

This research contributes to three UN Sustainable Development Goals. With respect to SDG 9 (Industry, Innovation, and Infrastructure), the study examines how startups can achieve digital sovereignty while maintaining technological competitiveness, thereby supporting the development of resilient infrastructure and innovative capabilities in the Nordic startup ecosystem. Regarding SDG 16 (Peace, Justice, and Strong Institutions), the analysis of GDPR compliance and data protection practices contributes to institutional frameworks that protect individual rights in the digital age. With respect to SDG 12 (Responsible Consumption and Production), the findings on transparent data collection practices support responsible business conduct in the digital economy.

1.7 Thesis Structure

This thesis is structured as follows. Chapter 2 establishes the theoretical, legal, and empirical foundations, reviewing the digital sovereignty literature, GDPR framework, third-party service dependencies, Privacy Nutrition Labels, Nordic context, and prior methodological approaches. Chapter 3 develops Digital Sovereignty Theory as the analytical lens, operationalizing it through Fries's layer model and GDPR framework, and develops the empirical hypotheses. Chapter 4 details the research design, data collection pipeline, weighted scoring systems, analytical methods, and validity and reliability checks. Chapter 5 presents the empirical findings, including descriptive statistics, primary regression results, decomposition analyses, and robustness checks. Chapter 6 interprets the findings through the lens of Digital Sovereignty Theory, discusses practical implications for startups and policymakers, and situates the results in the broader scholarly literature. Chapter 7 synthesizes the theoretical, methodological, and practical contributions, and discusses limitations.

Chapter 2

Literature Review

2.1 Practical and Industry Background

Nordic startups face a fundamental tension between technology adoption and digital autonomy. As digital services become essential for scalability and competitiveness, these startups must choose between dominant US-based technology platforms and maintaining control over their data governance. The EU's strategic emphasis on digital sovereignty frames this choice, aiming to ensure European actors retain autonomy, control, and resilience over their digital infrastructure and data (European Commission, 2022; Roberts, 2021).

2.1.1 Nordic Startup Ecosystem Context

The Nordic region includes Denmark, Sweden, and Finland as EU member states, along with Norway and Iceland as EEA members. This distinctive European sub-region features strong digital economies, high English proficiency, and early technology adoption (Autio & Rannikko, 2016).

The Nordic startup ecosystem has gained international recognition for its success in technology entrepreneurship. Autio and Rannikko (2016) analyze Nordic startups in the global digital economy, highlighting the factors behind the region's strong performance.

The Organisation for Economic Co-operation and Development (OECD) reports on Nordic digital transformation provide important macro-level context for understanding the region's technology adoption patterns (Organisation for Economic Co-Operation and Development, 2024). These reports highlight the advanced state of digitalization in the Nordic countries and the supportive policy environments that have enabled this progress.

Nordic Council digital strategy documents provide additional regional context, outlining the shared priorities and cooperative approaches that characterize Nordic technology policy (Nordic Council of Ministers, 2023).

Industry composition in the Nordic startup ecosystem is notably concentrated in information technology and software development. Prior studies indicate that a significant majority of Nordic startups operate in the IT and Software sectors, with a substantial proportion specifically identified as SaaS providers (Autio & Rannikko, 2016; Nassery et al., 2024; Organisation for Economic Co-Operation and Development, 2024). This concentration reflects the region's strengths in digital innovation and its focus on scalable, globally competitive technology solutions, while also highlighting the region's heavy reliance on digital infrastructure and third-party services.

2.1.2 Technology Adoption Patterns

Nordic startups exhibit a pronounced cloud-first orientation, with US-headquartered providers dominating early-stage infrastructure choices (Bauer et al., 2025; van der Vlist et al., 2024). Survey evidence and market data consistently show that AWS, Google Cloud, and Microsoft Azure collectively account for most cloud workloads among European Small and Medium-sized Enterprise (SME)s, and Nordic startups, characterized by small domestic markets and early-stage resource constraints, are particularly reliant on these platforms for scalability (Jeyaseelan, 2025; Ozu, 2024).

Industry composition shapes these patterns: SaaS and Information Technology (IT) startups, which constitute most Nordic technology ventures (Autio & Rannikko, 2016; Nassery et al., 2024), typically integrate analytics, monitoring, and Artificial Intelligence (AI) services from day one of product development, often defaulting to US tools with extensive developer ecosystems. Healthcare and fintech startups face additional pressures, as domain-specific compliance requirements (e.g., HIPAA-adjacent analytics tools, financial-grade monitoring) are predominantly offered by US providers.

Regulatory awareness has grown since Schrems II (2020), but adoption of EU-based alternatives remains limited. Kroon et al. (2025) and Ruokonen (2025) document the disconnect between Nordic policymakers' sovereignty ambitions and startup-level technology decisions, attributing it to cost differentials, ecosystem maturity gaps, and the absence of regulatory incentives for domestic sourcing. This gap between policy aspiration and operational practice forms the empirical problem motivating this study.

2.1.3 Regulatory Environment

The Nordic region implements the GDPR across all five countries, with some national variations in application and enforcement.

Each Nordic country has a dedicated supervisory authority: Denmark and Norway share the name Datatilsynet; Sweden operates through Integritetsmyndigheten (IMY); Finland through the Office of the Data Protection Ombudsman (Tietosuoja); and Iceland through Persónuvernd. Norway and Iceland apply the GDPR via their EEA Agreement obligations rather than as EU member states.

Enforcement patterns across Nordic data protection authorities reveal both common approaches and national variations (Koskinen & Knaapi, 2025; Nouwens et al., 2025). While all five countries share the GDPR framework, differences in legal traditions, institutional capacities, and policy priorities can lead to variations in interpretation and enforcement. In comparative terms, the Swedish IMY and the Danish Datatilsynet have been the most active enforcers, issuing the largest fines and most detailed guidance, whereas the Finnish, Norwegian, and Icelandic authorities have pursued comparatively fewer high-profile actions, reflecting differences in market size and regulatory resources rather than divergent legal standards. Because cross-national variation in enforcement intensity could independently influence how cautiously startups handle data, country differences are examined separately and descriptively via one-way ANOVA prior to the regression analysis; the non-significant results (Chapter 5) support treating the Nordic sample as a single regulatory context.

2.2 Core Literature

2.2.1 Third-Party Service Dependencies and Cloud Ecosystems

The empirical study of third-party service dependencies has a two-decade history. Mayer and Mitchell (2012) established the foundational taxonomy of web-tracking practices, documenting how third-party scripts propagate across sites to build cross-context user profiles. Krishnamurthy and Wills (2009) showed that these ecosystems enable aggregation of data across multiple services, creating behavioral profiles that far exceed what any single provider collects directly. Subsequent large-scale measurement studies by Kashaf et al. (2020) and Kumar et al. (2023) confirmed that third-party dependencies are now structural features of the digital economy rather than optional add-ons, with ad-tech incentives creating persistent pressure toward data collection beyond what core functionality requires.

2.2.2 Cloud Service Adoption and Market Dynamics

Market dominance by AWS, Google Cloud, and Microsoft Azure has produced a concentrated cloud services landscape. These US-based companies provide critical infrastructure for much of the global digital economy, shaping US third-party service dependencies (Basa, 2024; Ozu, 2024; van der Vlist et al., 2024).

Vendor lock-in creates significant economic and technical barriers to switching cloud providers. Eisenmann et al. (2009) and Farrell and Saloner (1985) analyze how standardization and path dependence create lock-in effects that make provider changes difficult after commitment.

Thong (1999) and Grandon and Pearson (2004) reveal how SME adoption patterns create particular challenges for cloud service adoption and management. Startups, constrained by resources and needing rapid scalability, often rely heavily on major cloud providers despite potential long-term vendor lock-in costs (Anwar, 2025; Jeyaseelan, 2025).

Startup-specific studies reveal how new ventures balance scalable infrastructure needs with technology stack control (Bauer et al., 2025). Technical requirements, economic constraints, and market dynamics often lead startups to prioritize immediate functionality and cost-effectiveness over long-term strategic considerations.

2.2.3 Service Classification Frameworks

Classifying third-party services by function and jurisdictional origin is essential for operationalizing sovereignty risks in a measurable way. Fries et al. (2023) provide a foundational taxonomy that distinguishes between infrastructure-layer services (cloud hosting, Content Delivery Networks (CDNs)), data and AI services (analytics, machine learning), and marketing and advertising services, arguing that each layer carries distinct data-access and sovereignty implications. This layered approach allows researchers and practitioners to move beyond aggregate counts of third-party services toward a weighted assessment of dependency risk that reflects the different ways services interact with user data.

Prior work on third-party tracking (e.g., Kashaf et al. (2020) and Mayer and Mitchell (2012)) has documented service prevalence but has not systematically weighted these services by jurisdictional risk. Combining Fries et al.'s (2023) functional taxonomy with a jurisdictional lens (distinguishing US-controlled from EU-origin services) provides a more comprehensive framework for assessing sovereignty exposure. The full operationalization of this combined approach, including the specific weights applied in this study, is developed in Chapter 3.

2.2.4 Jurisdictional Conflicts and Legal Access

A second structural challenge stems from jurisdictional conflict between the GDPR and US surveillance legislation. The CLOUD Act (2018) allows US law enforcement to compel US-based providers to disclose data regardless of physical storage location, applying to EU subsidiaries of US-controlled entities through corporate control rather than data location (Christakis, 2019; European Data Protection Board & European Data Protection Supervisor, 2019; US Congress, 2018). The Schrems II ruling (C-311/18) explicitly found that Foreign Intelligence Surveillance Act (FISA) Section 702 and Executive Order 12333 fail to provide protections equivalent to EU fundamental rights, invalidating the Privacy Shield (Court of Justice of the European Union, 2020). The EDPB has concluded that a CLOUD Act compulsion order does not constitute a valid legal basis for data transfer under GDPR, and that no technical measure fully mitigates this surveillance risk (European Data Protection Board, 2020; European Data Protection Board & European Data Protection Supervisor, 2019).

This structural incompatibility between GDPR data minimization requirements and US legal access regimes creates dual compliance obligations for Nordic startups that use US-hosted services (Broeders et al., 2023; Mattoo & Meltzer, 2018). The full legal analysis and operationalization of jurisdictional risk in this study's scoring framework is developed in Chapter 3.

2.2.5 Privacy Nutrition Labels and Transparency Mechanisms

Privacy Nutrition Labels (PNLs) represent Apple's 2020 initiative to provide standardized, machine-readable disclosures of data collection practices for iOS apps (Apple Inc., 2020). This mechanism requires all apps in the App Store to disclose what data they collect and how that data is used, providing users with consistent information across all applications.

The PNL format uses a standardized set of data categories and a machine-readable JavaScript Object Notation (JSON) structure, making it possible to analyze data collection practices at scale. This standardization is particularly valuable for research, as it enables consistent comparison across large numbers of apps and organizations.

The primary purpose of PNLs is to enhance transparency for end-users, allowing them to make more informed decisions about which apps to use based on their privacy preferences. By providing consistent, comparable information about data collection practices, PNLs address the information asymmetry that has historically characterized the mobile app ecosystem (Apple Inc., 2024).

As of 2024, PNLs are mandatory for all iOS apps, with Apple providing guidance and tools to help developers accurately complete their disclosures (Apple Inc., 2024).

2.2.6 Empirical PNL Studies

Large-scale analyses of PNLs have provided important insights into data collection practices across the mobile app ecosystem. Y. Li et al. (2022) conducted an exploratory large-scale analysis of App Store data, examining patterns of data collection across different app categories and identifying trends in how developers disclose their practices.

Kollnig et al. (2022) studied the impact of Apple's App Tracking Transparency framework and PNLs on user privacy, finding that these mechanisms have increased awareness and enabled users to make more privacy-conscious choices.

Novović (2023) examined the relationship between PNLs and GDPR compliance, identifying both the benefits and the limitations of using standardized disclosures for regulatory purposes. This study highlights the potential for PNLs to support compliance efforts while also noting the challenges of relying on self-reported data.

Surma et al. (2024) assessed the integrity of Apple's privacy labels, examining whether PNLs accurately reflect actual data collection practices and whether they comply with GDPR requirements. This study provides important validation of the PNLs mechanism and identifies areas where improvements are needed.

Scoccia et al. (2022) conducted an empirical study of privacy labels on the Apple iOS Mobile App Store, providing one of the most comprehensive analyses of PNLs to date. Koch et al. (2022) complement this work by auditing whether developers' stated PNL disclosures are substantiated by actual app behaviour, finding systematic inaccuracies that underscore the limitations of self-reported labels as a compliance mechanism.

2.2.7 PNLs in Research Context

The advantages of using PNLs in research are significant. The machine-readable format enables automated analysis at scale, allowing researchers to examine patterns across thousands of apps efficiently. The standardized nature of PNLs ensures consistency in how data collection practices are described, making cross-app comparisons more reliable.

Standardization across all iOS apps means that researchers can develop consistent methodologies for analyzing data collection practices. This standardization also facilitates reproducibility, as other researchers can apply the same methods to the same dataset.

Scalability to large samples is another key advantage. With hundreds of thousands of apps in the App Store, PNLs provide a dataset of unprecedented size for studying privacy practices. This scale enables analyses that would be impractical with traditional privacy policy analysis methods.

However, PNLs also have important limitations and challenges. As self-reported data, PNLs are subject to potential inaccuracies, with developers possibly misrepresenting their data collection practices either intentionally or unintentionally. Cranor (2022) highlights the risk of gaming, where developers might strategically disclose information in ways that make their apps appear more privacy-friendly than they actually are. Empirically, Rodríguez et al. (2023) find 66.5% discrepancy rates between iOS and Google Play PNL disclosures for the same apps, a result that raises questions about both developer diligence and app store enforcement. Krämer (2024) further argues that the structural design of App Store privacy labels, which distinguishes first-party from third-party tracking, does not satisfy GDPR Articles 12–14 transparency standards and structurally advantages platform-owned apps over third-party developers by exempting first-party tracking from disclosure requirements.

Another significant limitation is that PNLs do not specify the legal basis for data collection, which is a crucial requirement under the GDPR (Bygrave & Tosoni, 2020; Georgieva & Kuner, 2020). Without information about whether data collection is based on consent, contract necessity, legitimate interest, or other legal bases, it is difficult to fully assess compliance with EU data protection requirements.

Validation studies are needed to assess the accuracy of PNLs and their correspondence with actual data collection practices (Bhusal et al., 2025; T. Li et al., 2022; Zhang et al., 2022). Such studies could involve comparing PNLs disclosures with detailed privacy policy analysis, code inspection, or network traffic monitoring to verify that the disclosed practices match actual behavior. Jain et al. (2023) make a methodologically important contribution in this direction, presenting the Atlas system which automatically detects discrepancies between privacy policies and privacy labels using natural language processing, finding non-trivial inconsistency rates that validate the concern that self-reported disclosures require cross-validation.

2.3 Digital Sovereignty Theory: Origins and Scope

The empirical phenomena documented above (US market dominance, vendor lock-in, and jurisdictional conflict) have been theorized under the umbrella of Digital Sovereignty Theory (Digital Sovereignty Theory (DST)). This section surveys the theory's intellectual origins, scope, and prior empirical applications to position the more detailed operationalization developed in Chapter 3.

2.3.1 Conceptual Origins

The concept of digital sovereignty draws on classical political theories of state autonomy while extending them to the domain of digital infrastructure and data governance (Couture, 2019; Pohle & Thiel, 2020). Early formulations, associated with the European political response to the National Security Agency (NSA) surveillance revelations of 2013, focused primarily on state-level control over communication networks and data flows (Broeders et al., 2023). Subsequent scholarship broadened the concept in two directions: downward, to the organizational level of firms and public institutions (Hummel et al., 2021), and upward, to the supranational level of regional blocs such as the EU (Bradford, 2020).

Pohle and Thiel (2020) provide the foundational conceptual synthesis used in this thesis, tracing digital sovereignty as the capacity of states and other actors to exercise authority over their digital infrastructure, data flows, and technological capabilities. Building on this foundation, Hummel et al. (2021), Bellanova (2022), and Szabelka (2025) identify three analytically distinct but empirically interrelated dimensions: *autonomy* (the capacity to make independent technological choices free from external coercion), *control* (the ability to ensure that data processing remains subject to domestic legal frameworks), and *resilience* (the capacity to withstand disruptions arising from foreign jurisdictional actions or commercial decisions), demonstrating their applicability to both state actors and private organizations.

2.3.2 EU Policy Instantiations

Digital Sovereignty Theory has been operationalized at the policy level through several EU initiatives. The European Data Strategy (2020) and the subsequent Data Governance Act signal the EU's intent to position European actors as autonomous participants in the global data economy rather than passive recipients of US-defined technical norms (European Commission, 2022). The Gaia-X initiative, a Franco-German federated cloud infrastructure project, is perhaps the most direct institutional expression of the resilience dimension: by creating EU-hosted, standards-compliant alternatives to AWS and Google Cloud, it aims to reduce the structural dependence that DST identifies as the primary sovereignty risk for European organizations (Blancato, 2023).

The GDPR itself functions as an instrument of digital sovereignty in the control dimension: by restricting international data transfers (Chapter V) and imposing strict conditions on processing by data controllers regardless of their geographic origin, it attempts to bring US-controlled services within the reach of EU legal requirements (Kuner, 2021). The limits of this reach, exposed by the Schrems II ruling (C-311/18) and the CLOUD Act, define the residual sovereignty gap that this thesis investigates empirically.

2.3.3 Prior Empirical Applications and Research Gap

Despite its growing policy prominence, DST has been tested empirically in only a handful of studies. Hummel et al. (2021) survey organizational data sovereignty practices in the German in-

dustrial context, finding that firms systematically underestimate jurisdictional risk when adopting cloud services. Garcia Caceres (2025) examine how EU SMEs navigate the CLOUD Act–GDPR tension in their procurement decisions, documenting substantial compliance uncertainty. Kroon et al. (2025) and Ruokonen (2025) provide Nordic-level policy analyses documenting the gap between sovereignty aspirations and observed technology adoption among Nordic firms.

What remains absent from the literature is a large-scale *quantitative* test of DST's core prediction (that higher US service dependency is positively associated with greater GDPR-weighted data sensitivity) in the startup context. Startups differ from the incumbent firms studied by Hummel et al. (2021) and Garcia Caceres (2025) in important ways: they operate under tighter resource constraints, make foundational technology choices at a formative stage, and may lack the legal counsel to assess jurisdictional risk systematically. This study addresses the gap by operationalizing DST's dimensions as observable, weighted constructs and testing the theory's predictions against data from 501 Nordic startups. The theoretical operationalization is developed in Chapter 3.

2.4 Competing Theoretical Perspectives

Before adopting DST as the primary theoretical lens, this section briefly considers three established alternatives and explains why each is less suited to the present research question.

Resource-Based View (RBV) Barney (1991) conceptualizes competitive advantage as arising from inimitable, valuable internal resources. Applied to technology adoption, the RBV would predict that startups adopt US services because these services grant access to capabilities (developer ecosystems, pre-trained models, global CDNs) that constitute temporary competitive advantages. While the RBV illuminates *why* startups adopt particular services, it does not address the *consequences* of adoption for data governance or jurisdictional exposure. The present research question concerns the relationship between service choices and data collection practices, which lies outside the RBV's explanatory scope.

Transaction Cost Economics (TCE) Williamson (1979) explains governance choices (markets versus hierarchies) by reference to asset specificity, uncertainty, and transaction frequency. Applied here, TCE would explain vendor lock-in as a consequence of asset-specific investments in US platforms. Like the RBV, TCE is a productive complement for understanding *adoption dynamics* but does not theorize sovereignty risk or data sensitivity as outcomes. It also operates at the level of bilateral transaction governance rather than multi-stakeholder regulatory conflict.

Institutional Theory DiMaggio and Powell (1983) and related neo-institutional accounts explain organizational behavior through coercive, normative, and mimetic pressures. In the startup context, mimetic isomorphism (the tendency to copy technology stacks of successful peers) may partly explain the observed concentration of US service usage. However, institutional theory predicts *homogenization* of practices and does not derive predictions about the quantitative relationship between service dependency intensity and data sensitivity, which is the specific empirical question this thesis investigates.

Digital Sovereignty Theory is preferred because it directly theorizes the relationship between external technology dependencies and outcomes in data governance and compliance exposure, generates a falsifiable directional prediction (higher US dependency $\Rightarrow$ higher data sensitivity), and maps onto the specific legal context of the GDPR–CLOUD Act conflict. The theoretical operationalization is developed in Chapter 3.

2.5 Summary of Literature

This literature review has established the empirical foundations for understanding the relationship between US third-party service dependencies and data collection practices among Nordic startups. This review has surveyed research on third-party service dependencies and cloud ecosystems, assessed PNLs as a transparency mechanism, and provided Nordic-specific context.

The literature establishes several key points that inform this study. First, empirical research demonstrates the pervasiveness of third-party service dependencies and the central role that US-based providers play in the contemporary digital economy. Second, PNLs provide a valuable standardized mechanism for studying data collection practices at scale. Third, the concept of data sovereignty, understood as the capacity of actors to exercise meaningful control over their data (Hummel et al., 2021), has emerged as a central lens for evaluating the consequences of technology dependencies, complementing Digital Sovereignty Theory's focus on infrastructure autonomy, jurisdictional control, and ecosystem resilience.

Despite the comprehensive literature on these individual topics, five gaps motivate the present study.

The most important is geographic: despite extensive research on global technology adoption (van der Vlist et al., 2024) and on the Nordic startup ecosystem as a whole (Autio & Rannikko, 2016; Organisation for Economic Co-Operation and Development, 2024), no quantitative study has specifically examined how Nordic startups navigate this tension. The most relevant Nordic-level evidence comes from policy analyses (Kroon et al., 2025; Ruokonen, 2025) that document the gap between sovereignty aspirations and practice but do not test it empirically. A related gap concerns the measurement instrument itself: while PNLs have been analysed at scale (Kollnig et al., 2022; Y. Li et al., 2022; Scoccia et al., 2022), their accuracy relative to actual data collection practices remains under-validated (Bhusal et al., 2025; T. Li et al., 2022; Zhang et al., 2022), limiting confidence in any single-source analysis that relies on them alone.

Two further gaps are methodological. First, although the scoring schemes developed here build on established legal and theoretical frameworks (Bygrave & Tosoni, 2020; Fries et al., 2023; Georgieva & Kuner, 2020), no prior study validates comparable weighted approaches to quantifying data sensitivity and service dependency risk in a single empirical model. Finally, the choice calculus behind service selection (why startups opt for US over EU alternatives beyond immediate cost and functionality) remains poorly documented at the startup level (Bauer et al., 2025; Jeyaseelan, 2025), as does the question of how these dependencies evolve as firms mature. Temporal and sector-specific dynamics remain almost entirely uncharted.

This study addresses these gaps through quantitative analysis of 501 Nordic startups, novel operationalization using Fries et al.'s (2023) layer model and jurisdictional lens, and weighted scoring for dependencies and data sensitivity. By connecting empirical analysis to theoretical foundations (Chapter 3), it assesses how US technology dependencies relate to data collection practices in one of Europe's most vibrant startup regions.

Chapter 3

Theoretical Framework

This chapter develops Digital Sovereignty Theory (DST) as the primary theoretical lens for the empirical analysis. Section 3.1 defines the theory and its three core dimensions. Section 3.2 operationalizes those dimensions into the weighted scoring systems used in the empirical analysis. Section 3.3 derives the study's testable hypotheses from the theoretical framework.

3.1 Digital Sovereignty Theory

3.1.1 Definition and Scope

Digital Sovereignty Theory, as articulated by Pohle and Thiel (2020), traces the emergence of digital sovereignty as a political concept understood as the capacity of states and other actors to exercise authority over their digital infrastructure, data, and technological capabilities. This conceptualization encompasses both state-level sovereignty and the autonomy of organizations and individuals within a digital ecosystem.

For startups specifically, DST highlights the importance of maintaining agency in technology selection and data governance. Startups face particular challenges here because they often lack the resources and market power of larger enterprises, making them more vulnerable to path dependence and vendor lock-in (Roberts, 2021).

3.1.2 Three Dimensions: Autonomy, Control, and Resilience

Digital Sovereignty Theory can be understood through three analytically distinct but empirically interrelated dimensions: autonomy, control, and resilience (Bellanova, 2022; Hummel et al., 2021; Szabelka, 2025). Together they provide a framework for assessing the degree to which an actor maintains sovereignty over its digital infrastructure and data.

Autonomy: independent technological choices Autonomy refers to the ability to select, adopt, and configure technology without external pressure or constraints. In the context of this study, autonomy is undermined when startups feel compelled to use particular US-based services due to market dominance, network effects, or vendor lock-in (David, 1985; Eisenmann et al., 2009).

Control: data governance under local jurisdiction Control refers to the ability to ensure that data processing complies with local laws and regulations, particularly the GDPR in the EU context. The use of US-based services creates significant challenges for maintaining control, as

these services may be subject to US legal requirements that conflict with EU data protection standards (Garcia Caceres, 2025; Kuner, 2021).

The Clarifying Lawful Overseas Use of Data Act (CLOUD Act, 2018) and Foreign Intelligence Surveillance Act (FISA) Section 702 represent particularly significant threats to control, as they enable US law enforcement to compel US-based providers to disclose data regardless of where that data is physically stored (European Data Protection Board & European Data Protection Supervisor, 2019; US Congress, 2018). Critically, the *Schrems II* ruling (C-311/18) explicitly found that US surveillance laws do not provide protections equivalent to EU fundamental rights, leading to the invalidation of the Privacy Shield (Court of Justice of the European Union, 2020).

When Nordic startups use US-controlled services, including EU subsidiaries of US parents, their data may be subject to these US legal frameworks, creating a fundamental conflict with GDPR requirements and undermining the startups' ability to maintain control over their data governance (Christakis, 2019; Michels et al., 2024). The EDPB has explicitly warned that processors subject to the CLOUD Act cannot provide sufficient guarantees of compliance with the GDPR (European Data Protection Board & European Data Protection Supervisor, 2019).

Resilience: withstanding disruptions Resilience refers to the ability to maintain operations despite foreign jurisdictional actions or other disruptions. Dependence on US services creates vulnerabilities for Nordic startups, as these dependencies represent single points of failure that could be disrupted by legal actions, service changes, or other events outside the startups' control (Bellanova, 2022).

3.2 Operationalization

The GDPR functions not only as a compliance requirement for Nordic startups but also as a concrete instrument for operationalizing Digital Sovereignty Theory. Its risk-based architecture establishes a legally authoritative hierarchy of data sensitivity, provides an explicit framework for classifying cross-border data transfers by jurisdiction, and imposes proportionality obligations that map directly onto the autonomy, control, and resilience dimensions of DST (Bygrave & Tosoni, 2020; Georgieva & Kuner, 2020; Hummel et al., 2021). By grounding the weighted scoring systems in GDPR provisions (specifically Articles 4, 6, 9, 32, and 44), this framework bridges the gap between abstract sovereignty theory and measurable empirical constructs, ensuring that the operationalization carries legal validity alongside theoretical coherence (Moniz, 2024).

3.2.1 Service Layer Classification and Dependency Scoring

Fries et al.'s (2023) layer model provides a concrete framework for applying Digital Sovereignty Theory to the classification of third-party services. This model categorizes services into three layers based on their function and potential sovereignty implications, with higher layers representing greater sovereignty risk.

Infrastructure layer (×3 weight) The infrastructure layer includes cloud hosting, content delivery networks, and server providers that provide the foundational infrastructure on which other services depend. These services have the highest sovereignty risk because they involve direct control over data storage and processing.

Examples of infrastructure services include AWS, Google Cloud, Azure, and Heroku. The jurisdictional concern with these services is particularly acute given the application of the

US CLOUD Act, which can require US-based companies to provide data to US authorities regardless of where the data is physically located.

In this operationalization, infrastructure services receive a weight of ×3, reflecting their high sovereignty risk. This weighting reflects that these services provide the foundation for other digital operations and that control over them is essential for maintaining digital sovereignty.

Data and AI services layer (×2 weight) The data and AI services layer includes analytics platforms, machine learning services, and data processing tools that require data access for their functionality. These services often involve data flows to the service provider's infrastructure, creating potential sovereignty concerns.

Examples include Google Analytics, Firebase, Amplitude, and Mixpanel. The jurisdictional concern here is that data may be processed or stored in locations that subject it to foreign legal regimes, potentially conflicting with EU data protection requirements.

Data and AI services receive a weight of ×2 in this operationalization, reflecting their moderate sovereignty risk. While these services do not provide the foundational infrastructure of the first layer, they still involve significant data access and processing that can create sovereignty concerns.

Marketing and analytics layer (×1 weight) The marketing and analytics layer includes advertising platforms, customer relationship management tools, and other services that often use aggregated or anonymized data. Examples include Google Ads, Meta Ads, and HubSpot.

While these services typically have lower sovereignty risk than the other layers, they are still subject to foreign jurisdiction and may involve data collection practices that have privacy implications. Marketing and analytics services receive a weight of ×1 in this operationalization.

Weighted dependency score Figure 3.1 synthesizes how the three service layers translate into sovereignty risk. The logic proceeds in two steps. First, per-layer counts of US-classified services are multiplied by their respective weights and summed into a single weighted dependency score:

$$S = \sum_{\ell \in \mathcal{L}} c_\ell \cdot w_\ell$$

where $\mathcal{L} = \{\text{Infrastructure, Data \& AI, Marketing}\}$, c_ℓ is the count of US-classified services detected in layer ℓ for a given startup, and $w_\ell \in \{3, 2, 1\}$ is the layer weight. The weights reflect differential data-access intensity: infrastructure services receive $w = 3$ because cloud providers hold the most comprehensive and legally compellable access to startup data; data and AI services receive $w = 2$ because their operation requires continuous data flows; and marketing services receive $w = 1$ because their data access, while real, is narrower in scope. Second, the aggregated score simultaneously erodes all three DST dimensions through overlapping legal and market mechanisms, including CLOUD Act exposure, GDPR–US jurisdictional conflict, and vendor lock-in, none of which maps cleanly onto a single dimension. A startup compelled to disclose data under the CLOUD Act loses control (data leaves EU legal protection), resilience (operations may be disrupted), and autonomy (the compulsion constrains independent decision-making). The dependency score therefore functions as a single composite indicator of aggregate sovereignty erosion.

3.2.2 Data Sensitivity Classification

The GDPR's risk-based approach establishes a legally authoritative hierarchy of data sensitivity that directly grounds the weighted data collection scoring system used in this study (Bygrave

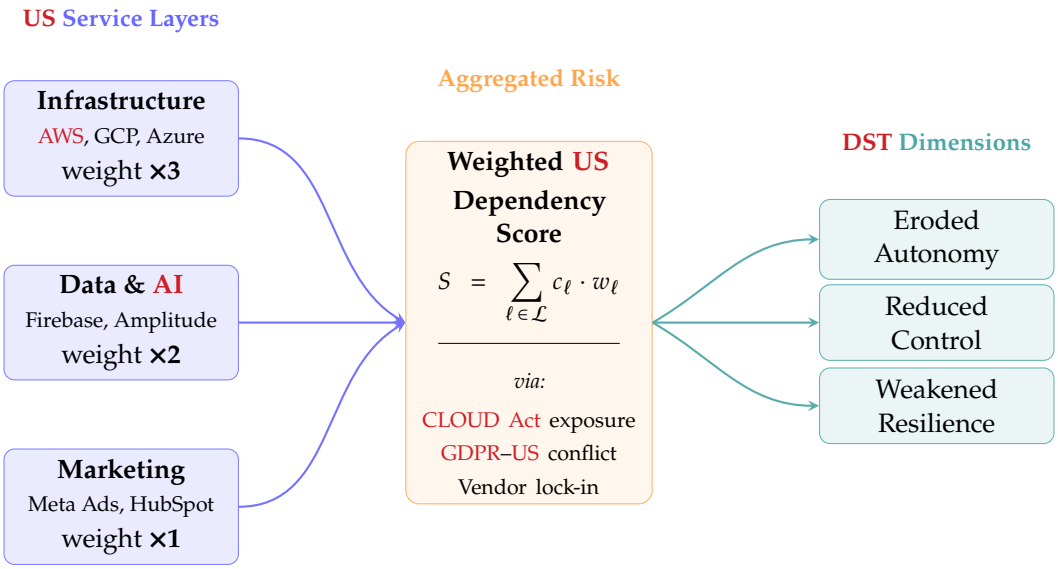


Figure 3.1: Sovereignty-Risk Framework.

& Tosoni, 2020; Georgieva & Kuner, 2020; Moniz, 2024).

Sensitive personal data (×4 weight) Article 9 of the GDPR identifies special categories of personal data that require explicit consent or strict conditions for processing (Georgieva & Kuner, 2020). These include:

- Health data
- Biometric data
- Genetic data
- Racial or ethnic origin
- Political opinions
- Religious or philosophical beliefs
- Trade union membership
- Data concerning a person's sex life or sexual orientation

The ×4 weight for sensitive personal data reflects its highest level of protection under the GDPR and its potential for discrimination or other harmful uses.

General personal data (×3 weight) Article 4 of the GDPR defines personal data broadly as "any information relating to an identified or identifiable natural person" (Bygrave & Tosoni, 2020). This includes direct identifiers (names, email addresses, physical addresses, phone numbers) and indirect identifiers (Internet Protocol (IP) addresses, device Identifiers (IDs)).

General personal data can be processed under various legal bases established in Article 6, including consent, contract necessity, and legitimate interest. These data types require standard protection measures but do not have the same stringent requirements as Article 9 data.

The ×3 weight for general personal data reflects its standard level of protection under the GDPR and its importance for privacy and data governance.

Usage and behavior data (×2 weight) Usage and behavior data (*e.g.* browsing history, clickstream data, app usage patterns) are *analytical categories* rather than explicitly defined GDPR legal classifications. They reflect processing contexts rather than legal data types. This data

is typically processed under the legitimate interest legal basis (Article 6(1)(f)), which requires balancing tests and privacy impact assessments.

The ×2 weight for usage and behavior data reflects its moderate level of protection under the GDPR and its importance for understanding user behavior and preferences.

Technical or diagnostic data (×1 weight) Technical or diagnostic data (*e.g.* device type, operating system version, crash logs, performance metrics) are *analytical categories*, not explicitly defined in GDPR. They reflect processing contexts rather than legal classifications. These data types have minimal privacy impact and are often processed under legitimate interest with minimal additional requirements.

The ×1 weight for technical or diagnostic data reflects its lowest level of protection under the GDPR and its minimal privacy impact.

3.2.3 Jurisdictional Classification

The GDPR's approach to cross-border data transfers provides a crucial framework for operationalizing jurisdictional concerns in this study. Article 44 restricts data transfers to third countries without adequacy decisions, and the *Schrems II* ruling (2020) has strengthened the requirements for mechanisms such as Standard Contractual Clausess (SCCs) (European Data Protection Board, 2021; Kuner, 2021; Svantesson, 2017).

The fundamental conflict between the GDPR and the CLOUD Act provides the conceptual basis for the jurisdictional operationalization in this study. The CLOUD Act allows US law enforcement to compel US-based providers to disclose data regardless of where that data is physically stored, while the GDPR seeks to protect the privacy rights of EU citizens and restrict cross-border data transfers (European Data Protection Board & European Data Protection Supervisor, 2019; US Congress, 2018).

Moreover, the US does not have a history of considering foreign or international law in surveillance matters; instead, it prioritizes domestic legal authorities even when they conflict with foreign jurisdictions (Deeks, 2016; Parrish, 2023). This structural imbalance, in which US law applies extraterritorially but EU law does not reciprocally apply to US data, erodes EU digital sovereignty by creating asymmetrical legal risk (Chung, 2026; Cristiano & Monsees, 2025).

The operationalization addresses this conflict by classifying service providers based on the jurisdiction of their parent company's headquarters (US, EU, or Other). The layer weights established in Section 3.2.1 apply to all services regardless of jurisdiction; separate weighted dependency scores are then computed for each jurisdictional group using the same weight structure. This allows direct comparison between US and non-US dependencies. The primary hypothesis concerns the US dependency score specifically, as US-headquartered services are uniquely subject to CLOUD Act and FISA Section 702 compulsion regardless of where data is physically stored, an extraterritorial legal reach that has no equivalent for EU- or Other-headquartered providers.

While the primary analysis uses layer-type weights without additional jurisdictional multipliers, two complementary robustness checks confirm that the main findings hold: a binary dependency classification (US=1.0, Non-US=0.0) that treats all non-US services equivalently (Appendix B.6), and re-estimation under three alternative data sensitivity weighting schemes that leave the standardized US dependency effect virtually unchanged (Appendix D.1.4).

3.3 Hypothesis Development

3.3.1 Mechanistic Pathways

Three theoretically motivated pathways may account for a positive association between US service dependencies and data collection practices, should such an association be empirically observed. These are propositions derived from theory and prior qualitative evidence; they are not independently tested in this thesis, which focuses on the aggregate relationship between dependency and sensitivity scores.

Technical pathway: service design and data requirements Many US-based analytics, personalization, and AI services are architecturally designed around continuous data ingestion: analytics services require detailed usage streams to compute meaningful metrics, personalization engines require user profiles to deliver ranked recommendations, and AI services require training corpora (Zuboff, 2022). This architecture may create structural pressure for startups to expose more user data to the service as a condition of receiving its full functionality, thereby expanding reported data collection practices.

Contractual pathway: data processing obligations and terms of service Terms of service and GDPR-mandated Data Processing Agreements (Data Processing Agreements (DPAs)) with US providers may specify data-sharing obligations as a precondition of service use (Roberts, 2021). These contractual structures can broaden the categories of data that a startup formally collects and transfers, thereby inflating reported data sensitivity scores even if the startup itself has limited data governance ambitions.

Legal pathway: jurisdictional exposure and compliance incentives Jurisdictional conflict between the CLOUD Act, FISA Section 702, and the GDPR may create indirect compliance incentives to maintain granular records (Garcia Caceres, 2025; US Congress, 2018). Startups integrating US services must navigate dual compliance obligations, and the practical response may be broader data documentation and collection. This pathway is theoretically plausible but empirically difficult to isolate from the technical and contractual pathways; the cross-sectional design does not allow for causal attribution.

These three pathways are not mutually exclusive and may reinforce one another. They motivate the positive direction of H1 while acknowledging the observational limits of the present study.

3.3.2 Theoretical Prediction and Hypotheses

The theoretical building blocks introduced above, namely DST's three dimensions and Fries et al.'s (2023) weighted layer model, translate into a concrete, falsifiable empirical prediction. Each additional US-headquartered service a startup integrates increases its exposure to CLOUD Act and FISA Section 702 compulsion. Because high-dependency-weight services (infrastructure, data and AI) also impose the greatest data-ingestion requirements by design, a startup's weighted US dependency score is expected to co-vary positively with the sensitivity of data it collects and discloses in its PNL.

DST therefore predicts a positive association between US third-party service dependencies and data collection practices: as startups become more dependent on US services, their weighted data collection scores should increase, reflecting the collection of more extensive and sensitive data. The following hypotheses formalize this prediction:

H1 (alternative hypothesis): Nordic startups with higher dependence on US third-party services exhibit higher GDPR-weighted data sensitivity scores ($\beta > 0$, $p < 0.05$). This prediction follows from the three mechanistic pathways identified above: US services impose data collection demands through their technical design, grant broad data processing rights via contractual terms, and create legal compliance incentives through the extraterritorial reach of US surveillance law (Garcia Caceres, 2025; Roberts, 2021; US Congress, 2018). In this operationalization, the independent variable is the US Dependency Weighted Score and the dependent variable is the GDPR-Weighted Data Sensitivity Score.

H0 (null hypothesis): There is no statistically significant association between US third-party service dependency and GDPR-weighted data sensitivity scores ($\beta = 0$). Failure to reject H0 would indicate that the choice of service jurisdiction does not predict data collection intensity, contrary to the predictions of Digital Sovereignty Theory.

3.4 Scope and Limitations

The theoretical framework developed here is bounded in scope: it operationalizes digital sovereignty at the organizational level of analysis and does not address state-level governance dynamics or macro-geopolitical dimensions of EU–US digital relations (Broeders et al., 2023; Hummel et al., 2021). Empirical limitations arising from the cross-sectional design, reliance on self-reported PNLs, and iOS-only sample are discussed in the context of the empirical findings in Chapter 7.

Chapter 4

Methodology

4.1 Research Philosophy

This study adopts a *positivist* epistemological stance and *objectivist* ontology. In this view, social and technical phenomena have an existence independent of the observer (objectivism), and knowledge about them is best derived from observable, measurable evidence (positivism). The study assumes an objective reality in which the relationship between US third-party dependencies and data collection practices can be measured empirically through observable phenomena. This aligns with established quantitative Information Systems research traditions (Maier et al., 2023; Orlikowski, 1992).

4.2 Research Approach

This study employs a *deductive* research approach, meaning it begins from established theory and tests hypotheses derived from it against empirical data, rather than building theory up from observations. The hypotheses are derived from Digital Sovereignty Theory (Pohle & Thiel, 2020) and are stated formally in Chapter 3, Section 3.3.2.

4.3 Research Design

This study quantitatively examines the relationship between US third-party dependencies and data collection practices among Nordic startups. Following Information Systems best practices (Maier et al., 2023), it uses a cross-sectional design, capturing 501 Nordic startups at a single point in time (2 to 4 May 2026). This design identifies patterns and correlations but cannot infer causality.

The study collects no primary human subject data. All data came from public sources: startup metadata from thehub.io, PNLs from the Apple App Store, and privacy policies from startup websites. No Institutional Review Board (IRB) or ethics approval was therefore required, consistent with ethical frameworks for Internet research (Association of Internet Researchers, 2012; Fiesler et al., 2019).

4.3.1 Unit of Analysis

The unit of analysis is the individual startup, not mobile applications, countries, or sectors. The sample of 501 Nordic startups with iOS apps and public privacy policies represents digital startups across Denmark, Sweden, Finland, Norway, and Iceland, all subject to both EU data protection regulations and global technology market realities.

This unit permits examination of how startup characteristics (size, age, industry, business model, and funding) shape US third-party service adoption and data collection, following established Information Systems research practices (Maier et al., 2023).

4.4 Data Collection

Data were collected from 2–4 May 2026 using a multi-source approach, grounded in established web scraping methods (Brown et al., 2025) and ethical frameworks for Internet research (Association of Internet Researchers, 2012; boyd danah & Crawford, 2010; Niman & Rich, 2011).

The operationalization of Digital Sovereignty Theory into measurable constructs, including the weighted scoring systems for service dependencies and data sensitivity, is detailed in Chapter 3: dependency scoring in Section 3.2.1 and data sensitivity classification in Section 3.2.2. The analytical approach employing Ordinary Least Squares (OLS) regression with robustness checks is described in Section 4.5.8 below.

4.4.1 Sample Construction Pipeline

Sample construction followed a four-step pipeline to ensure comprehensive coverage of Nordic startups while maintaining data quality. Step-level detail and the source and selection logic at each stage are provided in Appendix B.1.

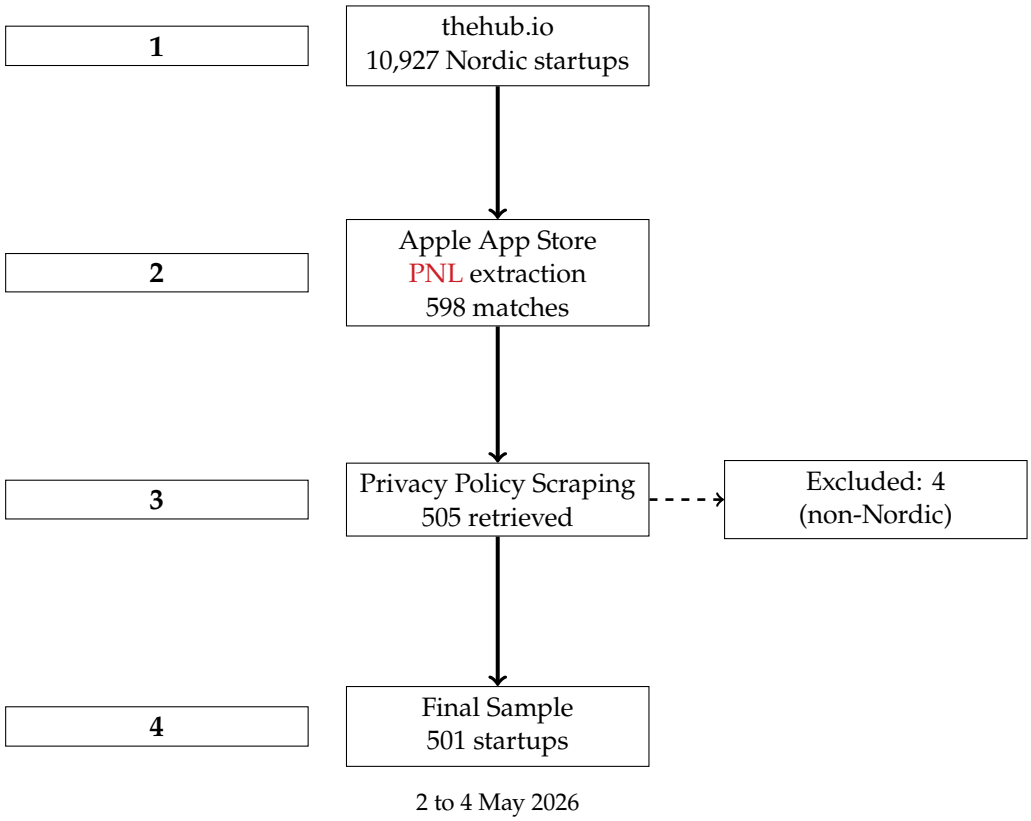


Figure 4.1: Sample Construction Pipeline

4.4.2 Final Sample Characteristics

The final sample includes 501 startups across five Nordic countries. By location as listed on thehub.io: Denmark (42.7%), Sweden (31.7%), Finland (16.0%), Norway (16.2%), and Iceland

Table 4.1: Sample Construction Pipeline Steps

Step	Source	N
1	thehub.io	10,927
2	Apple App Store	598
3	Privacy Policies	505
4	Location Filtering	501

(0.6%). Percentages sum to more than 100% because 27 startups (5.4%) list locations across multiple Nordic countries and are counted once per listed location. Of these 27, nineteen span two Nordic countries, seven span three, and one spans four, producing 36 additional country appearances in total (and thus a sum of 107.2%). Each startup enters the regression as a single observation. Country-level analyses (Table 5.1) use the startup's first-listed location only.

The sample consists primarily of digital startups, with industry concentration in IT and Software (51.5%) and SaaS (40.1%). Most startups are small enterprises: 50.5% employ 1–10 people, 34.5% employ 11–50, and 7.2% employ 51–100.

4.4.3 Data Sources

The data collection strategy employed a multi-source approach that leverages established web scraping methodologies (Brown et al., 2025; Luscombe et al., 2021) to systematically gather comprehensive information about Nordic startups and their data practices. Each source contributes a distinct component of the dataset: organizational characteristics, data collection practices, and third-party service dependencies are drawn from separate sources and linked at the startup level.

Startup metadata was obtained from thehub.io, a comprehensive platform that provides detailed information about entrepreneurial ventures across the Nordic region. This source supplied essential organizational characteristics including company names, geographic locations, website URLs, industry classifications, organizational size metrics, and funding status information.

Privacy Nutrition Labels data was systematically collected from the Apple App Store in JSON format. These standardized disclosures, which Apple introduced in 2020 as a requirement for all App Store applications (Kollnig et al., 2022; Novović, 2023; Surma et al., 2024), provide machine-readable information about data collection practices across 32 distinct data categories.

Public privacy policies were collected directly from startup websites to identify third-party service dependencies. These documents, through which companies fulfil their transparency obligations under GDPR Articles 13 and 14, contain detailed disclosures about the external services that startups utilize for various functions including infrastructure provision, data analytics, marketing, and customer engagement.

4.4.4 Methodological Validation

PNL processing follows methods established by Y. Li et al. (2022), Scoccia et al. (2022), and Khandelwal et al. (2023) for large-scale privacy label analysis. The accuracy concerns raised by Koch et al. (2022) and Jain et al. (2023), who find systematic discrepancies between self-reported PNL data and actual app behaviour, apply to this study: the dependent variable relies solely on PNL disclosures and is not cross-validated against code-level or network-traffic data. This limitation is acknowledged in Chapter 7.

For privacy policy analysis, the study addresses the same analytical challenge as Wilson et al. (2018), Zimmeck et al. (2019), and the POLIGRAPH system (Cui et al., 2023): identifying and classifying third-party services from unstructured text. The method here uses an Large Language Model (LLM)-based approach in place of the rule-based Natural Language Processing (NLP) methods those systems employ. The extraction pipeline is documented in Appendix F.

The extraction pipeline runs *Mistral Large 2512* (`mistral-large-2512`) three times independently on each privacy policy, following the self-consistency approach of Zhu et al. (2025) and Pitre et al. (2025); services identified by ≥ 2 of the three runs are retained. This consensus threshold reduces idiosyncratic extraction errors while preserving recall for services mentioned in varied natural-language phrasings. As noted in Appendix F.4, individual run-level outputs were not persistently logged, so a formal inter-agent agreement rate cannot be computed post hoc.

The theoretical derivation and legal grounding of both weighting schemes are presented in Chapter 3 (Sections 3.2.1 and 3.2.2); the schemes are tabulated with sensitivity analyses in Appendix B.3.

4.5 Data Analysis

4.5.1 Operationalization and Scoring System

The analysis operationalizes three variable categories: the dependent variable (data sensitivity score), independent variables (third-party dependency scores), and control variables.

4.5.2 Dependent Variable: Data Sensitivity

The dependent variable measures the sensitivity of data collected by each startup, derived from PNLs extracted from the Apple App Store (Brown et al., 2025; Kollnig et al., 2022; Novović, 2023).

Weighting scheme (GDPR-based) Sensitive personal data (Article 9 special categories: health, biometric, genetic, racial or ethnic origin, political opinions, religious beliefs, trade union membership, and data concerning sex life) receive the highest weight of ×4. General personal data (Article 4: names, email addresses, physical addresses, phone numbers, IP addresses, and device IDs) receive a weight of ×3. Usage and behavior data (browsing history, clickstream, app usage patterns, search history) receive ×2. Technical or diagnostic data (device type, Operating System (OS) version, crash logs, performance metrics) receive ×1.

This risk-based hierarchy aligns with GDPR's emphasis on proportionality, where more sensitive data requires stronger protections and more stringent legal requirements (Bygrave & Tosoni, 2020; Georgieva & Kuner, 2020; Moniz, 2024).

Data sensitivity score The dependent variable is a GDPR-weighted data sensitivity score calculated as:

$$\begin{aligned}\text{Data Sensitivity Score} = {} & 4 \times \text{Sensitive Personal Data} + 3 \times \text{General Personal Data} \\ & + 2 \times \text{Usage and Behavior Data} + 1 \times \text{Technical or Diagnostic Data}\end{aligned}$$

where each term denotes the number of data categories of that type that a startup collects, as disclosed in its PNL, and the integer coefficients (4, 3, 2, 1) are the GDPR-based sensitivity weights defined in the weighting scheme above.

The resulting index produces a continuous-like score (mean = 24.82, observed range = 0–192) that represents the overall data sensitivity of a startup's data collection practices. This continuous nature makes OLS regression with heteroskedasticity-robust standard errors (HC1) the appropriate primary specification for the analysis.

4.5.3 Independent Variables: Third-Party Dependency Scores

The independent variables measure dependence on third-party services, identified from privacy policies and classified by jurisdiction and type. The theoretical rationale and weight derivation are in Chapter 3 (Section 3.2.1).

Weighting scheme (Fries layer model) Three service tiers are defined with increasing sovereignty risk: Infrastructure services (cloud hosting, CDNs, server providers; weight ×3), Data and AI services (analytics, ML, data processing; weight ×2), and Marketing and Analytics services (advertising, CRM, customer support; weight ×1). Services are classified by parent-company headquarters jurisdiction (US, EU, or Other).

Dependency score For each startup and each jurisdiction $j \in \{\text{US}, \text{EU}, \text{Other}\}$, the weighted dependency score is:

$$S_j = \sum_{\ell \in \mathcal{L}} c_{j,\ell} \cdot w_\ell$$

where $c_{j,\ell}$ is the count of jurisdiction-j services in layer ℓ and $w_\ell \in \{3, 2, 1\}$ is the layer weight. The primary independent variable is S_{US}; the combined Non-US score ($S_{\text{EU}} + S_{\text{Other}}$) enters the model as a control.

4.5.4 Control Variables

The models include five organizational characteristics as controls for potential confounders of the dependency–sensitivity relationship.

Industry (entered as dummy variables for SaaS, Healthcare and Life Science, Service, and Media and Entertainment, with the remaining categories as the reference) controls for sector-specific data requirements. Fintech and Healthcare startups, for instance, operate under domain-specific compliance regimes that independently raise both service complexity and data sensitivity.

Company size (number of employees) controls for resource availability: larger startups command greater engineering capacity and legal counsel, which may independently drive both service complexity and data governance sophistication.

Company age (years since founding) controls for organizational maturity, which may shape both accumulated technology stacks and established data practices.

Business model (entered as dummy variables for B2C, B2G, and Marketplace, with B2B as the reference) controls for customer-facing differences in data collection: consumer- and marketplace-oriented startups typically gather more behavioral and personal data than business-facing peers.

Funding stage (Pre-seed, Seed, Series A, Series B+, IPO, with Bootstrap as the reference) serves as a proxy for investor-driven growth pressure, which plausibly influences how aggressively a startup integrates external services and how much data it collects to demonstrate growth metrics.

Together these five controls allow assessment of whether the US dependency–data sensitivity association is a general feature of the ecosystem or an artifact of specific industry, size, age,

business-model, or maturity profiles. Cross-national variation is examined separately and descriptively (one-way ANOVA by country in Section 5.2) rather than as a regression covariate, because country effects on the weighted scores are non-significant and the Nordic sample is treated as a single regulatory context under the shared GDPR framework.

4.5.5 Data Processing and Classification

Raw PNLs and privacy policy data were processed into structured datasets for analysis.

4.5.6 PNL Processing

Apple provides PNLs as machine-readable JSON, which makes extraction straightforward. All 32 data categories were parsed from each app's label, each category mapped to its tier in the four-level GDPR-based weighting scheme (Sensitive ×4, General Personal ×3, Usage ×2, Technical ×1) using a fixed crosswalk table, and each startup's weighted data sensitivity score calculated as the sum of (category count × weight) across all tiers. The crosswalk was established before data collection and applied without deviation. Apple's PNL requirement has been mandatory for all App Store apps since 2020 (Apple Inc., 2020, 2024).

4.5.7 Privacy Policy Processing

Privacy policy processing was more complex due to the unstructured nature of these documents. The study employed a multi-agent LLM consensus approach with three independent agents per policy:

Multi-agent workflow

1. Each agent independently read the privacy policy and extracted all mentioned third-party services
2. Service names were normalized (*e.g.*, "Google Ireland Limited", "Google LLC", "Alphabet Inc." all mapped to "Google")
3. Services were classified by region (US, EU, Other) based on parent company headquarters
4. Services were classified by type (Infrastructure, Data and AI, Marketing and Analytics) using Fries et al.'s (2023) layer model
5. Consensus validation: Services identified by ≥ 2 agents were retained; inconsistencies flagged for manual review

Service name normalization Deterministic string matching algorithms handled common variations. For example, all variations of Google's corporate entities were normalized to "Google" while maintaining distinctions between service families (*e.g.*, "Google Analytics" vs. "Google Ads" vs. "Google Cloud"). This ensures consistent classification across all startups.

Region and type lookup Standardized lookup tables mapped each normalized service name to its parent company headquarters jurisdiction and service type category. These tables were populated using publicly available information from company websites, regulatory filings (Securities and Exchange Commission (SEC), EU transparency registers), and industry directories (Crunchbase, LinkedIn). Ambiguous cases were verified manually.

4.5.8 Analytical Approach

The relationship between US third-party service dependencies and data collection practices is analyzed using OLS regression as the primary specification, with a Poisson count model as a robustness check (pre-specified as Negative Binomial; the optimizer did not converge and Poisson is the equivalent fallback at the observed dispersion level; see Poisson specification below). The GDPR-weighted data sensitivity score is a composite measure for which OLS with HC1 robust standard errors provides valid inference, as established in Section 4.5.2. As a robustness check, the relationship is re-estimated on unweighted count data (the count of GDPR data-type tiers collected regressed on the count of US service categories used) with a Poisson count model, to test whether the findings depend on the weighted operationalization of the variables.

4.5.9 Software and Tools

The analysis was conducted using Python 3.11 with key libraries including pandas 2.x, numpy, statsmodels, scipy, seaborn, and matplotlib. The complete software stack, including version details and reproducibility information, is provided in Appendix B.2.

4.5.10 Regression Models

Multiple regression specifications were estimated to test the hypotheses and assess robustness. The primary specification is OLS regression with heteroskedasticity-robust standard errors (HC1), with a Poisson count-model robustness check (pre-specified as Negative Binomial) and additional specifications.

Primary OLS specification

$$\text{Data Sensitivity}_i = \beta_0 + \beta_1(\text{US Dependency}_i) + \beta_2(\text{Controls}_i) + \epsilon_i$$

where $\text{Data Sensitivity}_i$ is the GDPR-weighted data sensitivity score for startup i, US Dependency_i is its weighted US third-party dependency score, Controls_i is the vector of industry, size, age, business-model, and funding-stage controls, β_0 is the intercept, β_1 and β_2 are the estimated coefficients, and ϵ_i is the error term. The same notation conventions apply to the remaining specifications below.

The primary model employs OLS regression with heteroskedasticity-robust standard errors (HC1), which is statistically appropriate for the dependent variable. While the underlying components are counts, the weighted sum (4 Sensitive + 3 Personal + 2 Usage + 1 Technical) creates a continuous-like variable where OLS provides valid inference for assessing associations. This model directly tests H1 by estimating the coefficient on US Dependency while controlling for industry, size, age, business model, and funding stage. The OLS specification provides interpretable coefficients and appropriate standard errors for inference, following best practices in Information Systems research for analyzing weighted composite outcomes (Maier et al., 2023; Wooldridge, 2019).

Log-transformed OLS specification

$$\log(\text{Data Sensitivity}_i + 1) = \beta_0 + \beta_1 \log(\text{US Dependency}_i + 1) + \beta_2(\text{Controls}_i) + \epsilon_i$$

Natural log transformation of both dependent and independent variables addresses potential skewness. A constant of 1 was added before transformation to handle zero values (Curto,

2023; Russell, 2000). This specification provides an alternative parameterization to test for non-linear relationships.

Poisson count-model robustness check (unweighted data; pre-specified as Negative Binomial, optimizer did not converge)

$$\log(\mathrm{E}[\text{Data Type Count}_i]) = \beta_0 + \beta_1(\text{US Service Count}_i) + \beta_2(\text{Controls}_i)$$

As a robustness check, the relationship is re-estimated on unweighted count data. The dependent variable is the count of GDPR data-type tiers each startup collects (0–4: Sensitive, General Personal, Usage, Technical), the primary independent variable is the count of US service categories used (0–3: Infrastructure, Data and AI, Marketing), and the controls are identical to the primary OLS model. In practice, the Negative Binomial optimizer did not converge across four estimation methods (BFGS, Nelder-Mead, Newton, and default), and Poisson regression was used as the fallback estimator. This is inconsequential for inference. Because the data-type count is not over-dispersed (variance-to-mean ratio ≈ 0.75), the Negative Binomial dispersion parameter would collapse toward zero and the estimator reduces to its Poisson limit in any case. The reported coefficients use heteroskedasticity-robust (HC1) standard errors.

This count-model robustness check ensures the findings are not an artifact of the weighted operationalization of the dependent and independent variables.

Interaction specification

$$\begin{aligned}\text{Data Sensitivity}_i = \beta_0 &+ \beta_1(\text{US Dependency}_i) + \beta_2(\text{Service Type}_i) \\ &+ \beta_3(\text{US Dependency}_i \times \text{Service Type}_i) + \beta_4(\text{Controls}_i) + \epsilon_i\end{aligned}$$

This OLS interaction model tests whether the effect of US dependencies varies by service type (Infrastructure, Data and AI, Marketing). Service Type is a categorical variable with three levels. The interaction term probes whether the relationship differs across service categories.

Service-type sub-analyses Separate OLS regressions were estimated for each service type to identify which category drives the observed relationship: Three sub-analyses are estimated, each using one service-type count as the key independent variable: the US Infrastructure service count (unweighted), the US Data and AI service count (unweighted), and the US Marketing service count (unweighted), respectively. Using raw counts rather than weighted scores allows each service type's raw functional relationship with data sensitivity to be estimated independently of the tier-weight assumptions applied in the primary model.

All sub-analyses include the same control variables as the primary OLS specification and use OLS regression with heteroskedasticity-robust standard errors.

4.5.11 Statistical Assumptions and Diagnostics

Key statistical assumptions were verified; one diagnostic flag is disclosed below:

OLS assumptions (primary model)

- Dependent variable nature: the dependent variable is a GDPR-weighted data sensitivity score (mean = 24.82, observed range = 0–192). As discussed in Section 4.5.2, this composite score supports valid inference under OLS with HC1 robust standard errors; the diagnostic checks below confirm the choice.

- Independence: the Durbin-Watson test, which detects serial correlation in regression residuals, returns d = 2.05 (values near 2 indicate no autocorrelation), as expected with cross-sectional data where each observation represents a different startup.
- Homoskedasticity: the Breusch-Pagan test failed to reject constant residual variance (LM = 19.606, p = 0.5463), indicating no substantial heteroskedasticity. HC1 robust standard errors are used throughout regardless, as they are asymptotically valid whether or not heteroskedasticity is present and are standard practice in cross-sectional regression (Wooldridge, 2019).
- Normality: the Shapiro-Wilk test, which assesses whether residuals are normally distributed, rejects normality ($p < 0.001$) for OLS residuals. However, with N=501, the Central Limit Theorem supports the approximate normality of the sampling distribution of coefficient estimates.
- Functional form: the Ramsey RESET test detects potential misspecification (F = 15.440, p = 0.0004), indicating possible omitted non-linear terms or interaction effects. The primary OLS specification is nonetheless retained: both the log-transformed OLS and the Poisson count-model robustness check produce qualitatively consistent findings ($\beta > 0$, $p < 0.05$), suggesting the RESET signal reflects omitted confounders or distributional heterogeneity rather than a fundamental misspecification of the linear relationship between US dependency and data sensitivity. The most direct robustness response to the RESET signal is the log-transformed OLS specification, which addresses the right-skewed distribution of the dependent variable, the most plausible source of the nonlinearity the RESET test detects. It returns $\beta = 0.199$, $p < 0.001$ for US dependency, qualitatively identical to the primary specification, confirming that the RESET flag does not undermine the core finding.

  Full OLS diagnostic test results are in Appendix E.

Count-model assumptions (robustness check)

- Operationalization: the robustness check uses unweighted counts, specifically the count of GDPR data-type tiers collected (0–4) regressed on the count of US service categories used (0–3), with the same controls as the primary model.
- Dispersion: the data-type count is not over-dispersed (variance-to-mean ratio ≈ 0.75), so the Negative Binomial dispersion parameter collapses toward zero and the estimator reduces to its Poisson limit, reported with HC1 robust standard errors.
- Model fit: Pseudo $R^2 = 0.026$, Log-Likelihood = −853.5, AIC = 1743.0.

Model selection rationale The primary specification uses OLS regression with HC1 robust standard errors, appropriate for the GDPR-weighted sensitivity score (Section 4.5.2); robustness is confirmed by the Poisson count-model check described above.

4.6 Research Quality

4.6.1 Reliability

Classification consistency was ensured through multiple mechanisms:

Automated normalization Rule-based service name normalization used deterministic string matching algorithms. Corporate entity variants were collapsed to canonical names: 'Google Ireland Limited', 'Google LLC', and 'Alphabet Inc.' all mapped to 'Google'; 'Amazon Web Services' and 'Amazon.com' to 'AWS'; and 'Meta Platforms', 'Facebook', and 'Instagram' to 'Meta'.

Standardized lookup tables Region and type assignments used predefined tables mapping normalized service names to jurisdiction (US: parent headquartered in the USA; EU: parent headquartered in the EU/EEA; Other: all remaining jurisdictions) and service type (Infrastructure, Data and AI, or Marketing and Analytics).

Multi-agent consensus validation Three independent LLM agents reviewed each privacy policy. Services identified by ≥ 2 agents were retained with high confidence. Inconsistencies (disagreements between agents) were flagged for manual review and resolved through consensus discussion.

This multi-agent approach builds on established automated privacy policy analysis methods (Bui et al., 2021; Cui et al., 2023; Hosseini et al., 2020; Rahman et al., 2022) while incorporating recent advances in LLM systems (Zhu et al., 2025).

Limitation: inter-agent agreement rate not reported The run-level outputs from each of the three agent instances were not persistently logged during data collection, making it impossible to calculate a formal inter-agent agreement rate (e.g., Cohen's κ or percentage agreement) post hoc. Future replications of this methodology should log per-agent extractions before the consensus step so that classification reliability can be quantified. This is acknowledged as a methodological limitation and discussed further in Section 7.4.

4.6.2 Validity

Construct validity The weighting schemes are grounded in established legal and theoretical frameworks. Data sensitivity weights follow the GDPR's explicit legal hierarchy (Article 9 special categories = ×4, Article 4 general data = ×3). Dependency scores follow Fries et al.'s (2023) layer model, which classifies services by their strategic importance to digital sovereignty (Bygrave & Tosoni, 2020; Fries et al., 2023).

Convergent validity checks Convergence with external indicators was assessed. Data Sensitivity scores correlated strongly with the number of distinct Apple PNL data categories reported out of 14 ($r = 0.81$, $p < 0.001$), confirming that the weighted composite tracks collection breadth as expected. Industry group means showed directional consistency with expectations (Healthcare: 27.4, Fintech/Commerce: 29.6, Software/Tech: 20.3), though a one-way Analysis of Variance (ANOVA) across the five broadest industry groupings did not reach statistical significance ($F(4, 496) = 2.23$, $p = .065$).

External validity The sample of 501 Nordic startups captures regional diversity across all five Nordic countries and multiple industries. However, the sample is limited to startups with iOS applications, excluding Android-only and web-only ventures; the Nordic regulatory context may not generalize to other regions; and the cross-sectional design limits temporal generalizability.

Internal validity Cross-sectional design limits causal inference. However, robustness checks across four model specifications strengthen confidence in observed associations. The consistency of results across OLS, log-transformed, interaction, and sub-analysis models suggests the observed relationship is not an artifact of model specification.

4.7 Ethical Considerations

This study was conducted following established ethical frameworks for public data research (Buck, 2021; Fiesler et al., 2019; Krotov, 2020).

All data were collected from publicly accessible commercial platforms (thehub.io, Apple App Store, startup websites) without authentication, subscription, or interaction with individual users. The study focused exclusively on organizational data (startup characteristics and service disclosures) rather than individual user data; no personally identifiable information was retained. Because the study collected no personal identifiers and did not interact with individuals, it is exempt from human subjects research requirements, consistent with frameworks from Buck (2021) and Fiesler et al. (2019). Data collection was limited to what was necessary to answer the research question, data were stored on encrypted local devices accessible only to the author, and collection dates (2–4 May 2026), sources, methods, and processing pipelines are fully documented.

4.7.1 Declaration on the Use of AI Tools

This study used AI tools in two distinct ways: as a documented research instrument and as a writing aid.

First, a large language model was employed as a core analytical instrument. The multi-agent consensus framework based on *Mistral Large 2512* was used to extract and classify third-party services from unstructured privacy policy text, as described in Section 4.4.4 and documented in full, including the prompt template and consensus rules, in Appendix F. This constitutes a methodological component of the study rather than writing assistance, and its outputs were subject to the consensus threshold and validation procedures detailed above.

Second, generative AI tools were used to assist with the writing process, specifically for drafting and restructuring some passages and for language editing such as grammar, phrasing, and clarity. All AI-assisted text was critically reviewed, fact-checked, and edited by the author, who verified every citation and factual claim against the underlying sources. No analytical results, statistical findings, or interpretations were generated by these tools. The author takes full responsibility for the entire content of this thesis, including any text produced with AI assistance.

With the data, operationalization, and validation procedures established, the empirical analysis proceeds as follows. Chapter 5 presents the empirical findings, beginning with patterns of US service dependency among Nordic startups, then testing the hypotheses using weighted regression models. Chapter 6 interprets these findings through the lens of *Digital Sovereignty Theory* and discusses their policy implications for the Nordic startup ecosystem.

Chapter 5

Results

5.1 Empirical Data Presentation

Variable definitions, scoring methodologies, and data sources are documented in the *Data Dictionary* (Appendix A).

5.2 Descriptive Statistics: Service Dependency Patterns

Nordic startups exhibit strong structural dependency on US services, using 3.64 US services on average (median: 2) versus 0.53 EU services (median: 0), a 6.8× difference ($t(500) = 17.88$, $p < 0.001$; Table 5.3). Most strikingly, 75.2% of startups use no EU services at all, indicating minimal adoption of European alternatives. This finding suggests that despite the EU's emphasis on digital sovereignty and the availability of European service providers, most Nordic startups continue to rely heavily on US providers for their technology needs.

Country-level variations exist within the Nordic region but are modest and not statistically significant across weighted scores. Table 5.1 presents mean US dependency and data sensitivity scores by country. Norway shows the numerically highest mean US dependency (8.5) and Sweden the lowest (6.0), yet a one-way ANOVA yields no significant country effect on either weighted score ($F(4, 496) = 1.27$, $p = .281$ for US dependency; $F(4, 496) = 0.97$, $p = .423$ for data sensitivity). A pairwise Norway–Sweden comparison on unweighted US service-category counts does reach significance (Welch's $t(119.1) = 3.01$, $p = .003$), suggesting marginal differences in service breadth that the weighting procedure attenuates. These non-significant country-level results indicate that structural US dependency is a Nordic-wide phenomenon rather than a country-specific outlier.

Industry variations show a similar pattern. Table 5.2 shows that data sensitivity scores vary numerically across sectors, with Consumer Goods, Healthcare & Life Science, and Fintech startups exhibiting the highest mean scores (31.8, 29.3, and 27.0 respectively) and SaaS and Education startups the lowest (18.1 and 17.9), but a one-way ANOVA does not reach significance ($F(8, 492) = 1.79$, $p = .077$). US dependency scores likewise do not differ significantly across industries ($F(8, 492) = 1.80$, $p = .075$). The absence of significant industry effects on either weighted score indicates that structural US dependency cuts across sectors rather than concentrating in particular business models, reinforcing the regression evidence in Section 5.3 that the dependency–sensitivity association holds net of industry controls.

This dominance is concentrated among a few providers: the top 10 US services account for 50.6% of all US service usage, with Google (53.5% adoption), Facebook (37.7%), and Google Analytics (27.1%) most prevalent. EU alternatives show minimal adoption (0.4–3.6% vs 6.6–53.5% for US counterparts) (Table 5.4).

Table 5.1: Mean Weighted US Dependency and Data Sensitivity by Country (N=501 Nordic Startups)

Country	N	Mean US Dependency	Mean Data Sensitivity
Denmark	214	6.40	22.85
Sweden	145	5.98	28.32
Finland	72	7.10	24.10
Norway	67	8.51	24.55
Iceland	3	5.67	20.00

Country assigned by location as listed on thehub.io; for startups listing multiple locations, the first-listed location is used. One-way ANOVA confirms no significant country effect on either score (US dependency: $F(4, 496) = 1.27$, $p = .281$; data sensitivity: $F(4, 496) = 0.97$, $p = .423$).

Table 5.2: Mean Weighted US Dependency and Data Sensitivity by Industry (N=501 Nordic Startups)

Industry	N	Mean US Dependency	Mean Data Sensitivity
Consumer Goods	51	5.61	31.75
Healthcare & Life Science	63	5.38	29.29
Other	52	7.37	28.65
Fintech	28	6.82	26.96
Media & Entertainment	50	9.40	26.02
IT & Software	133	6.05	23.34
Energy & Greentech	21	4.62	22.43
SaaS	57	6.28	18.05
Education	46	8.80	17.87

Note: Industry assigned by primary (first-listed) industry; categories with fewer than 20 startups are grouped as "Other." Rows ordered by mean data sensitivity.
One-way ANOVA: data sensitivity $F(8, 492) = 1.79$, $p = .077$; US dependency $F(8, 492) = 1.80$, $p = .075$. Neither weighted score differs significantly across industries at $\alpha = .05$.

Table 5.3: Descriptive Statistics: Third-Party Service Usage (N=501)

	Mean	Median	SD
US Services	3.64	2	4.2
EU Services	0.53	0	1.3
Non-US Services (EU+Other)	0.69	0	1.5

Paired t-test (US vs. EU counts per startup): t(500) = 17.88, p < 0.001
75.2% of startups use zero EU services
Non-US = EU services + Other region services

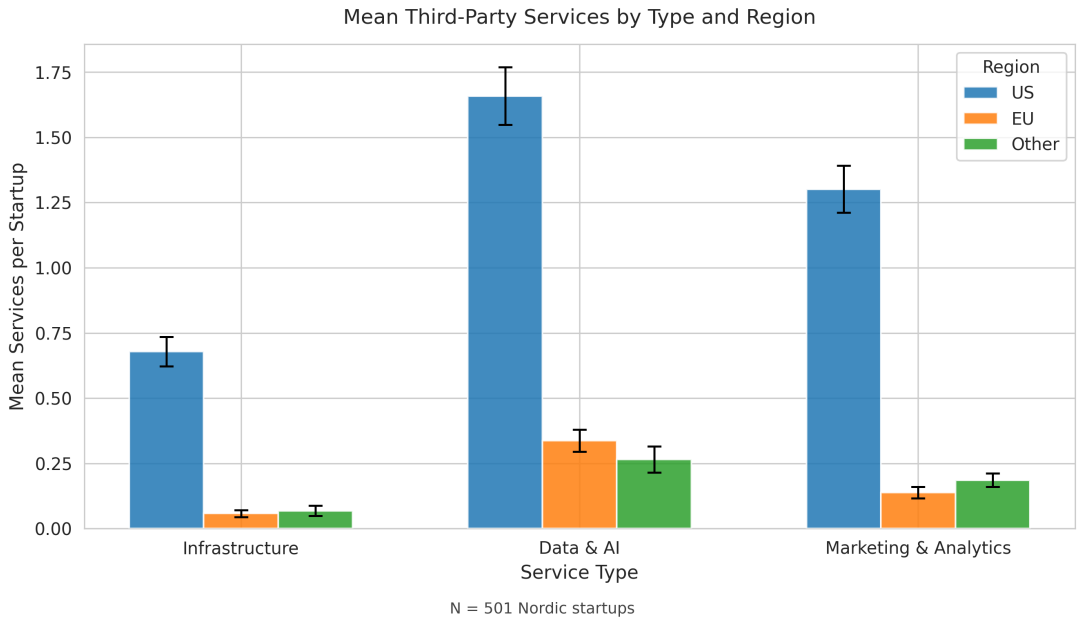


Figure 5.1: Mean third-party services per startup by type and region (±1 SE; N=501).

Table 5.4: Top 10 US and EU Services by Adoption (N=501 Nordic Startups)

Service	Startups	Percentage
1. Google	268	53.5%
2. Facebook	189	37.7%
3. Google Analytics	136	27.1%
4. LinkedIn	76	15.2%
5. Stripe	68	13.6%
6. Apple	59	11.8%
7. HubSpot	47	9.4%
8. Amazon Web Services	36	7.2%
9. Instagram	34	6.8%
10. YouTube	33	6.6%
Top 10 EU Services		
1. Hotjar	18	3.6%
2. Iubenda	5	1.0%
3. Cookiebot	4	0.8%
4. Pipedrive	3	0.6%
5. Brevo	3	0.6%
6. Typeform	3	0.6%
7. Crispchat	2	0.4%
8. Soundcloud	2	0.4%
9. Adyen	2	0.4%
10. Hetzner Online Gmbh	2	0.4%

% = share of 501 Nordic startups using that service. Top 10 US services account for 50.6% of all US service usage.

5.3 Primary Regression Results

The primary specification is OLS regression with HC1 robust standard errors, which is statistically appropriate for the GDPR-weighted data sensitivity score; the full statistical justification is in Section 4.5.2.

Table 5.5: Primary OLS Regression: GDPR-Weighted Data Sensitivity on US Dependency and Controls (N=501)

	Coef.	SE	t	p	95% CI	Std. β
Primary predictors						
US_Dependency_Weighted	0.571***	0.136	4.200	<0.001	[0.305, 0.838]	0.173
NonUS_Weighted	−0.473	0.362	−1.306	0.191	[−1.181, 0.236]	−0.070
Firm characteristics						
Employees_Num	0.082**	0.031	2.664	0.008	[0.022, 0.142]	0.195
Company_Age	−0.215	0.267	−0.806	0.420	[−0.738, 0.308]	−0.037
Industry (ref: Other)						
Industry_SaaS	−3.835	2.140	−1.792	0.073	[−8.028, 0.359]	−0.071
Industry_Healthcare_Life_Science	3.463	3.183	1.088	0.277	[−2.775, 9.702]	0.049
Industry_Service	1.026	3.542	0.290	0.772	[−5.917, 7.968]	0.014
Industry_Media_Entertainment	−1.989	3.236	−0.615	0.539	[−8.331, 4.354]	−0.026
Business model (ref: B2B)						
BusinessModel_B2C	8.854***	2.611	3.391	<0.001	[3.736, 13.971]	0.156
BusinessModel_B2G	0.538	4.482	0.120	0.905	[−8.248, 9.323]	0.004
BusinessModel_Marketplace	16.776**	5.124	3.274	0.001	[6.734, 26.818]	0.183
Funding stage (ref: Bootstrap)						
FundingState_Pre_seed	5.250	3.833	1.370	0.171	[−2.263, 12.762]	–
FundingState_Seed	7.030*	3.110	2.260	0.024	[0.934, 13.126]	–
FundingState_Series_A	6.076	3.936	1.544	0.123	[−1.639, 13.791]	–
FundingState_Series_Bplus	13.292	7.110	1.870	0.062	[−0.643, 27.227]	–
FundingState_IPO	23.957	13.602	1.761	0.078	[−2.703, 50.616]	–
FundingState_Undisclosed	14.188***	3.455	4.106	<0.001	[7.404, 20.972]	–
Intercept	8.912*	3.501	2.545	0.011	[2.049, 15.774]	–

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. HC1 heteroskedasticity-robust standard errors.
$R^2 = 0.180$, Adj. $R^2 = 0.151$, $N = 501$. Std. β not reported for dummy variables (–).

The core finding is that US dependency significantly predicts higher data sensitivity ($\beta = 0.571$, $p < 0.001$; std $\beta = 0.173$). A one-unit increase in the weighted US dependency score is associated with a 0.571-point increase in the GDPR-weighted data sensitivity score (on a scale where the mean is 24.82), holding all else constant.

For OLS regression, coefficients represent changes in the original units of the dependent variable. The standardized coefficient ($\beta = 0.173$) indicates that a one-standard-deviation increase in US dependency is associated with a 0.173 standard deviation increase in data sensitivity.

Non-US dependency shows no significant effect ($p = 0.191$), confirming that the association is specific to US-headquartered services rather than a general third-party dependency effect.

Practical effect size To ground the coefficient in substantive terms: the interquartile range of the weighted US dependency score is approximately 10 units (25th–75th percentile). Multiplying by $\hat{\beta} = 0.571$ yields a 5.71-point increase in GDPR-weighted data sensitivity across that range, equivalent to roughly 22% of one standard deviation of the sensitivity score ($\sigma \approx 26.5$). This represents a moderate and practically meaningful shift in data collection exposure, anal-

ogous to a startup moving from collecting primarily Technical Data to additionally collecting significant Usage and General Personal Data.

Control variable findings Several control variables reach significance alongside the primary predictor. Firm size (Employees) is a comparably strong predictor (std β = 0.195, p = 0.008), indicating that larger startups collect more sensitive data, likely reflecting greater product complexity and analytics capacity. Business model matters substantially: Marketplace firms carry the largest standardized coefficient (std β = 0.183, p = 0.001), and Business to Consumer (B2C) firms show a similar profile (std β = 0.156, p < 0.001). Both exceed the standardized US dependency effect (std β = 0.173), suggesting that consumer-facing business models are as powerful a predictor of data sensitivity as US infrastructure reliance. The Seed funding stage is modestly positive (p = 0.024), while Company Age, Business to Government (B2G), and industry dummies are non-significant, indicating that data collection intensity is shaped more by business model and operational scale than by firm maturity or sector per se. The FundingState (Undisclosed) dummy, covering the 151 startups (30.1% of the sample) whose funding stage was not disclosed in the thehub.io database, carries the largest funding-related coefficient in the model (β = 14.19, p < 0.001). This group's elevated data sensitivity likely reflects organizational maturity: undisclosed-stage startups have typically progressed beyond the MVP stage and developed more data-intensive products, yet have not (or choose not to) publicly announce their funding round. This finding also accounts for much of the Funding Stage block's 21.3% Shapley share (Section 5.3.1): once the Undisclosed category is set aside, the remaining stage dummies (Pre-seed, Series A, Series B+) are individually non-significant, suggesting that disclosed funding stages per se do not independently predict data sensitivity.

The model fit is acceptable with R^2 = 0.180 and Adj. R^2 = 0.151, explaining approximately 18% of the variance in data sensitivity scores. For cross-sectional social-science modelling, R^2 values of 10–20% are considered acceptable provided key predictors are statistically significant (Ozili, 2023), as is the case here. The unexplained variance reflects product-specific differences in functionality, the voluntary and self-reported nature of PNL disclosures (Krämer, 2024), and unmeasured organizational factors that a cross-sectional design cannot capture.



Figure 5.2: Weighted dependency score vs. GDPR-weighted data sensitivity score with LOESS trend lines, by jurisdiction (N=501).

5.3.1 Relative Importance of Predictors

Fig. 5.3 ranks every predictor on a common standardized scale with 95% confidence intervals. Firm size, US dependency, and the Marketplace and B2C business models are the significant positive predictors. Among the funding stage dummies, the Seed stage is modestly positive ($p = 0.024$) and the Undisclosed category is strongly positive ($p < 0.001$), reflecting the organizational maturity of startups that do not publicly disclose their funding round (discussed in Section 5.3). All industry dummies, Non-US dependency, company age, and the remaining funding stage controls are statistically indistinguishable from zero.

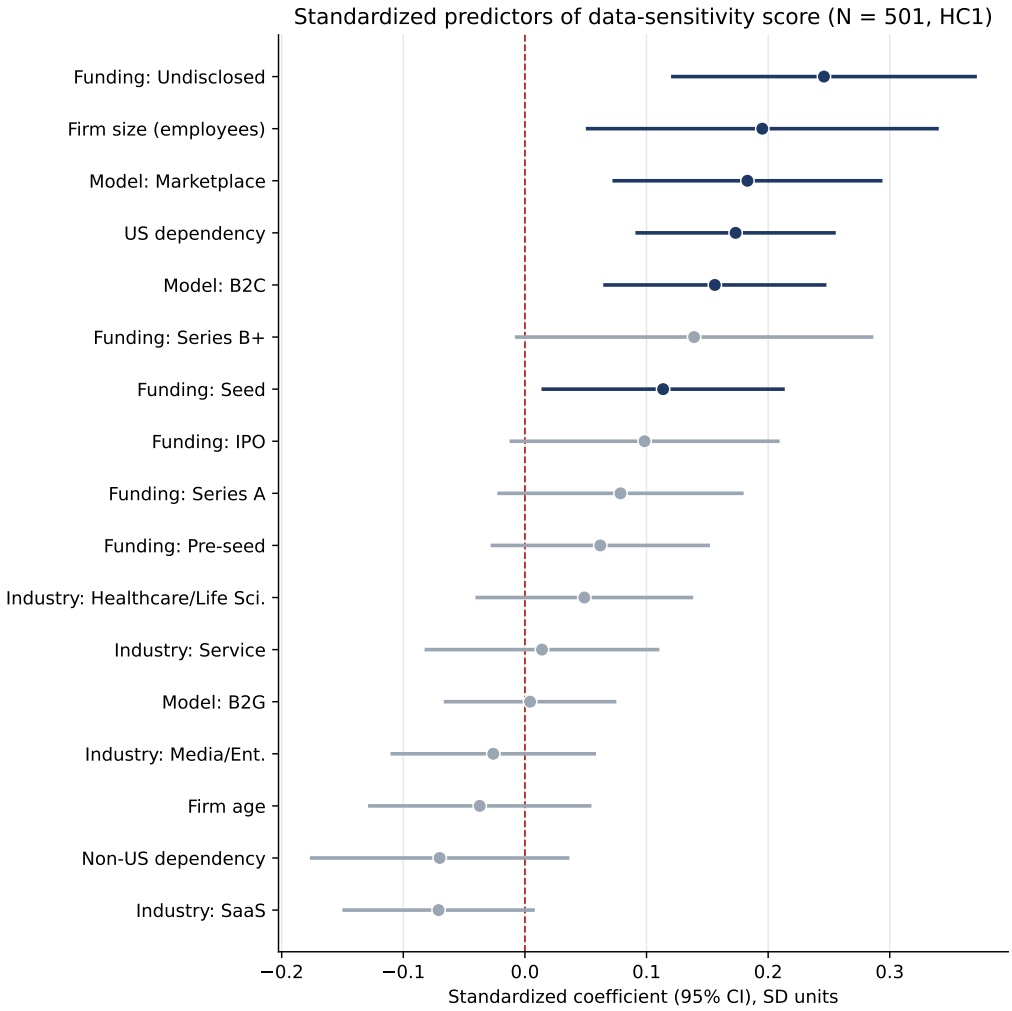


Figure 5.3: Standardized regression coefficients (95% confidence intervals, HC1 robust standard errors) for predictors of the GDPR-weighted data sensitivity score. Dark markers denote $p < 0.05$; the dashed line marks zero (N=501).

To quantify how much each predictor contributes to the explained variance, a Shapley decomposition of the model R^2 is computed exactly over all 2^7 subsets of seven predictor blocks (Table 5.6). US dependency ranks fourth of seven predictor blocks, accounting for 14.6% of explained variance. Business model (30.2%), firm size (24.2%), and funding stage (21.3%) each outrank it, with business model contributing more than twice as much. For funding stage, however, the 21.3% share should be interpreted with care: as noted in Section 5.3, the Funding Stage block's Shapley contribution is largely attributable to the FundingState (Undisclosed) category ($\beta = 14.19$, $p < 0.001$) rather than to the individually non-significant disclosed stage

dummies. Data collection intensity is therefore multi-causal: US dependency is one significant determinant among several rather than the singular driver, a reading consistent with the modest overall R^2.

Table 5.6: Relative Importance: Shapley Decomposition of Explained Variance (R^2) by Predictor Block (N=501)

Predictor block	Shapley R^2 share	% of explained variance
Business model	0.0545	30.2%
Firm size	0.0436	24.2%
Funding stage	0.0385	21.3%
US dependency	0.0264	14.6%
Industry	0.0139	7.7%
Non-US dependency	0.0019	1.0%
Firm age	0.0016	0.9%
Total	0.1803	100.0%

Shapley values computed exactly over all 2^7 subsets of the seven predictor blocks; shares sum to the full-model R^2.
US dependency ranks fourth of seven blocks (14.6%). Business model, firm size, and funding stage each account for more explained variance, consistent with a multi-causal account in which US dependency is a significant but not dominant driver.

5.4 What Drives the Effect?

5.4.1 By Service Type

US Data & AI services show the strongest association with data sensitivity by standardized effect size (std β = 0.131, β = 1.408, SE = 0.420, p < 0.001), followed by Infrastructure (std β = 0.107, β = 2.240, SE = 1.093, p = 0.041) and Marketing & Analytics (std β = 0.079, β = 1.036, SE = 0.596, p = 0.082) (Table 5.7). Note that US Infrastructure carries a larger raw coefficient (2.240 vs. 1.408) but acts on a narrower set of startups, yielding the smaller standardized effect. Each coefficient represents the change in the GDPR-weighted data sensitivity score associated with one additional service in that category, holding all controls constant.

Table 5.7: Regression: Data Sensitivity by Service Type (N=501)

	Coef.	Std. Err.	t	p	Std. Coef.
US Infrastructure	2.240*	1.093	2.05	0.041	0.107
US Data & AI	1.408***	0.420	3.36	<0.001	0.131
US Marketing	1.036†	0.596	1.74	0.082	0.079
EU Infrastructure	−3.231	3.531	−0.92	0.360	−0.037
EU Data and AI	2.356†	1.294	1.82	0.069	0.085
EU Marketing	0.729	1.832	0.40	0.691	0.013

Note: Each row represents a separate regression with data sensitivity as DV. Service type variables are unweighted counts.
Controls: Firm size, age, industry, business model, funding stage
*** p < 0.001, ** p < 0.01, * p < 0.05, † p < 0.10 (marginally significant)
Standardized coefficients (Std. Coef.) indicate effect sizes in standard deviation units (Cohen, 1988)

5.4.2 By Data Type

Decomposing by GDPR category reveals that US dependency associates most strongly with General Personal Data (+45% from Q1 to Q4, p = 0.004), Usage and Behavior Data (+65%, p = 0.002), and Technical Data (+38%, p = 0.016); see Table 5.9 notes for the percentage calculation formula. Notably, Sensitive Personal Data collection shows no association with US dependency (p = 0.477), suggesting that startups may be strategically avoiding legally restricted data categories under GDPR Article 9 while still expanding their collection of other data types.

This pattern is confirmed in a regression framework. Table 5.8 regresses each data-tier count separately on US dependency with the full control set: US dependency significantly predicts General Personal (std $\beta = 0.19$, $p < 0.001$) and Usage (std $\beta = 0.13$, $p = 0.002$) data, but shows no significant association with Sensitive Article 9 data ($p = 0.057$) or Technical data ($p = 0.097$). The convergence of the quartile test ($p = 0.477$) and the multivariate model isolates the Article 9 null as a robust feature of the data rather than an artifact of either method. Technical Data provides a contrasting illustration of how univariate and multivariate results can diverge: the quartile t-test yields a nominally significant association ($p = 0.016$), yet this effect does not survive the full control set in the regression model ($p = 0.097$). The difference indicates that Technical Data collection co-varies with confounders, most likely firm size and business model, rather than reflecting an independent relationship with US dependency.

Table 5.8: Strategic-Compliance Test: Association of US Dependency with Each Data Tier (N=501)

Data tier (GDPR)	Mean count	β_{US}	std err	p	std β
Sensitive (Art. 9)	0.46	0.0112	0.0059	0.057	0.077
General/Personal (Art. 4)	5.00	0.1267	0.0279	<0.001***	0.188
Usage & Behavior	2.98	0.0635	0.0203	0.002**	0.129
Technical/Diagnostic	2.00	0.0193	0.0116	0.097	0.070

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. HC1 robust SEs; each row is a separate OLS of the tier count on US dependency, Non-US dependency, and the full control set.
US dependency predicts collection of General/Personal and Usage data but shows *no* significant association with legally restricted Sensitive (Article 9) data, substantiating the strategic-compliance interpretation.

Table 5.9: Data Type Collection by US Dependency Quartile (N=501)

	Q1 (Low)	**Q2**	**Q3**	**Q4 (High)**
Sensitive Personal Data	0.4	0.3	0.7	0.5
General Personal Data	4.1	3.8	6.4	6.0
Usage and Behavior Data	2.1	2.4	4.2	3.5
Technical Data	1.6	1.8	2.5	2.2
Total Weighted Score	19.9	19.3	32.7	29.3

Q1 = lowest US dependency quartile, Q4 = highest; cells are mean counts per startup. Percentage changes in text computed as $(\bar{x}_{Q4} - \bar{x}_{Q1})/\bar{x}_{Q1}$. Non-monotone Weighted Score (Q3 > Q4) reflects within-group heterogeneity in service weighting; the regression uses the continuous dependency score and is unaffected.
Q1 vs. Q4 Welch t-tests: Sensitive $p = 0.477$; General Personal $p = 0.004$; Usage & Behavior $p = 0.002$; Technical $p = 0.016$.

This pattern, where non-sensitive data collection rises with US dependency while sensitive personal data collection remains flat, is discussed through the lens of Digital Sovereignty Theory in Chapter 6.

5.5 Robustness Checks

The primary finding demonstrates robustness across multiple model specifications. The positive association between US dependency and data sensitivity is consistent across OLS regression ($\beta = 0.571$, $p < 0.001$; std $\beta = 0.173$), log-transformed OLS ($\beta = 0.199$, $p < 0.001$; std $\beta = 0.162$), and a Poisson count model on unweighted data ($\beta = 0.0455$, $p = 0.048$; Incidence Rate Ratio (IRR) = 1.047; Appendix D.1.2). Pairwise interaction models test whether the effect of overall US dependency on data sensitivity varies by service type (full results in Appendix D.1.3). The Infrastructure interaction is significant ($\beta = -0.156$, $p = 0.018$), indicating that the marginal effect of each additional unit of US dependency is attenuated for startups that already rely heavily on US infrastructure providers. The Data and AI interaction is negative and directionally consistent with this pattern ($\beta = -0.060$) but does not reach conventional significance ($p = 0.077$) once controls are harmonized with the primary model. In the combined model including all three interaction terms, none reaches significance individually, which reflects multicollinearity among the correlated service-type counts. Jurisdictional robustness is confirmed using a binary service classification (Appendix B.6): replacing the continuous weighted score with a binary indicator (US = 1 if the startup uses any US-based services, 0 otherwise) yields $\beta = 6.807$ ($p = 0.005$; std $\beta = 0.118$), confirming that the association holds even at the coarsest possible operationalization of US dependency.

Model comparison The OLS primary model provides interpretable coefficients in the original scale. The Poisson robustness check confirms that the core association holds under a count-data estimator designed for discrete outcomes. The consistent direction and significance across both model specifications strengthens confidence in the findings. Full results are in Appendix D.1; the primary OLS model is Table 5.5 above.

Sensitivity to the weighting scheme Because the GDPR-weighted data sensitivity score relies on a chosen set of GDPR-based tier weights (4:3:2:1), the primary model is re-estimated under three alternatives: unweighted counts (1:1:1:1), a steeper Article 9 emphasis (8:4:2:1), and a shallower gradient (5:4:3:2). The US dependency coefficient stays positive and significant ($p < 0.001$) in every case, with a near-constant standardized effect (std $\beta = 0.16$–0.17) and stable fit ($R^2 = 0.17$–0.18) (Table D.5). The headline association is thus not an artifact of the particular weights chosen. Full results are in Appendix D.1.4.

5.6 Model Diagnostics and Limitations

Model diagnostics reveal several considerations relevant to model selection. The Shapiro-Wilk test rejects normality ($p < 0.001$) for OLS residuals, and the dependent variable exhibits 16.4% zeros while US dependency scores show 29.7% zeros. However, multicollinearity is low (max Variance Inflation Factor (VIF) = 3.15) and autocorrelation is absent (Durbin-Watson = 2.05).

Model selection These diagnostics are relevant for the OLS models. While the dependent variable does not exhibit perfect normality (Shapiro-Wilk p < 0.001), OLS with heteroskedasticity-robust standard errors remains valid for assessing associations at this sample size (N=501). The

consistency of results across OLS specifications and the Poisson robustness check (Section 5.5) strengthens confidence in the primary findings.

5.7 Chapter Summary

This study finds a statistically significant positive relationship between US third-party dependencies and GDPR-weighted data sensitivity scores among Nordic startups. The primary OLS regression specification reveals that each additional weighted US service is associated with a 0.571-point increase in the GDPR-weighted data sensitivity score ($\beta = 0.571$, $p < 0.001$; std $\beta = 0.173$). This relationship is consistently significant across multiple model specifications, including log-transformed OLS ($\beta = 0.199$, $p < 0.001$; std $\beta = 0.162$) and a Poisson count model on unweighted data ($\beta = 0.0455$, $p = 0.048$; IRR = 1.047), demonstrating robustness across different statistical approaches. The effect is primarily driven by Data and AI services and is characterized by increased collection of general personal data and usage data, while aligning with Digital Sovereignty Theory's predictions about the erosion of digital autonomy and control.

Weighted dependency distributions are shown in Appendix C.3, and US vs Non-US relationships in Appendix C.4. Industry-level differences are reported in Table 5.2 above.

Chapter 6

Discussion

This chapter interprets the empirical findings presented in Chapter 5 through the lens of Digital Sovereignty Theory, organized around the theory's three core dimensions: autonomy, control, and resilience.

6.1 Interpreting the Findings Through Theory

The empirical results provide strong support for Digital Sovereignty Theory (DST), showing that US third-party service dependencies are associated with patterns consistent with erosion of each of the theory's three core dimensions (Broeders et al., 2023; Garcia Caceres, 2025; Hummel et al., 2021). To make the theory falsifiable rather than merely illustrative, Table 6.1 states five DST predictions in advance and confronts each with the corresponding empirical result. Three predictions are supported, one is refined by a null result, and one holds only partially: dependency is a significant but non-dominant driver of data practices. This mixed scorecard, rather than a uniform confirmation, is what lends the theory empirical traction here.

6.1.1 Autonomy: The Erosion of Technological Choice

These findings are consistent with Digital Sovereignty Theory's prediction that external technology dependencies are associated with reduced autonomy. The structural imbalance documented in Section 5.2, where most Nordic startups rely exclusively on US services with negligible EU alternatives, indicates a severe narrowing of independent technological choice.

Mechanism: path dependence and lock-in effects The data amplify this picture. 75.2% of Nordic startups use zero EU services; 50.6% of all US service usage concentrates in just ten providers. Industry and country ANOVA tests are both non-significant ($F(8, 492) = 1.80$, $p = .075$ and $F(4, 496) = 1.27$, $p = .281$ respectively), confirming that this concentration is a structural feature of the ecosystem, not an artefact of any single sector or country. Path-dependence theory (David, 1985; Eisenmann et al., 2009; Farrell & Saloner, 1985) explains why: once startups commit to a dominant US platform, integrated ecosystems, proprietary Application Programming Interfaces (APIs), and data gravity effects make switching prohibitive.

Implication: market forces override sovereignty considerations The near-absence of EU service adoption indicates that cost, scalability, and ecosystem effects currently outweigh sovereignty considerations in startup technology decisions. Individual rational choices aggregate into collective dependence, a market failure that policy cannot correct by targeting individual startups.

Table 6.1: Digital Sovereignty Theory: Predictions versus Empirical Findings

DST prediction	Empirical result	Verdict
P1. Core association. Greater dependence on US providers is associated with more intensive (higher-sensitivity) data collection (H1: $\beta > 0$).	US dependency predicts the GDPR-weighted sensitivity score ($\beta = 0.571$, $p < 0.001$); H0 rejected. Positive and significant under every alternative weighting scheme (Table D.5).	Supported
P2. Jurisdictional specificity. The effect is specific to non-sovereign (US) dependency, not to third-party dependency in general.	Non-US dependency is not significantly associated with sensitivity ($\beta = -0.47$, $p = 0.191$); only US dependency loads.	Supported
P3. Layer gradient. Higher-sovereignty-risk layers (infrastructure, data & AI) drive the effect more than marketing.	US infrastructure (std $\beta = 0.11$) and data & AI (std $\beta = 0.13$, $p = 0.002$) are significant; US marketing is only marginal ($p = 0.058$) (Table 5.7).	Supported
P4. Erosion of control over sensitive data. If dependency erodes data control, collection of Article 9 special-category data should also rise.	No significant association with Article 9 data (regression $p = 0.057$; quartile test $p = 0.477$); the rise is confined to less-regulated tiers (Table 5.8).	Refuted[†]
P5. Dependency as a principal driver. Dependency should be a substantial determinant of data practices.	Significant but one driver among several: 14.6% of explained variance, below business model (30%) and firm size (24%) (Table 5.6).	Partial

[†] The Article 9 null refines rather than contradicts the theory: it indicates startups retain agency over legally restricted categories while dependency expands collection of less-regulated data, consistent with a strategic-compliance reading of digital sovereignty.

6.1.2 Control: Jurisdictional Conflicts and Data Governance

The association ($\beta = 0.571$, $p < 0.001$) is consistent with the three theoretically motivated pathways developed in Chapter 3 (Section 3.3.1): the *technical* pathway (service architectures associated with continuous data ingestion), the *contractual* pathway (DPA obligations associated with expanded disclosed data categories), and the *legal* pathway (dual-compliance pressures from CLOUD Act/FISA exposure (Kuner, 2021; Svantesson, 2017)). These remain interpretive framings; the cross-sectional design does not permit causal attribution to any single mechanism.

The jurisdictional specificity of the effect is itself informative. Non-US dependency (EU and Other services combined) shows no significant association with data sensitivity ($\beta = -0.473$, $p = 0.191$). This contrast isolates US legal jurisdiction, not third-party dependency in general, as the operative dimension. The Schrems II ruling (C-311/18) supplies the structural mechanism: US surveillance law does not meet EU fundamental rights standards (Court of Justice of the European Union, 2020), an asymmetry that EU-origin services do not share.

The data-type decomposition identifies *where* control erodes. US dependency predicts General Personal Data collection (Q1-to-Q4 increase: +45%, $p = 0.004$) and Usage and Behavior Data (+65%, $p = 0.002$), but shows no significant association with Article 9 sensitive data (regression $p = 0.057$; quartile test $p = 0.477$; Table 5.8). Control loss concentrates in the tiers that US analytics and AI platforms access by default, namely behavioral tracking and identity data, while startups retain agency over the most legally restricted categories. The erosion is partial and stratified, not uniform.

The service-type results identify *where in the stack* the loss is sharpest. US Data & AI services carry the strongest standardized effect (std $\beta = 0.131$, $p < 0.001$) and US Infrastructure the largest raw coefficient ($\beta = 2.240$, $p = 0.041$); US Marketing & Analytics reaches only marginal

significance ($p = 0.082$) (Table 5.7). This layered pattern aligns with the technical pathway: analytics and AI platforms access behavioral and identity data as a core product function, not a side effect of their adoption.

6.1.3 Resilience: The Risk of Single Points of Failure

The concentration of US service usage among just ten providers creates three resilience risks: technical (an AWS or Google Cloud outage cascades across the entire ecosystem simultaneously), legal (US court orders compel disclosure from all dependent startups at once, not only the primary target), and commercial (pricing changes or service deprecations reshape startup stacks with little warning). The infrastructure layer is the most acute single point of failure: 34.3% of startups use at least one US infrastructure service, and US Infrastructure carries the largest raw coefficient of any service type ($\beta = 2.240$), indicating deep integration per service rather than superficial adoption.

The negative relationship between US and Non-US dependency (Appendix C.4) confirms that startups concentrate rather than diversify their jurisdictional exposure. With only 0.53 EU-based services per startup on average, the ecosystem has minimal redundancy. EU initiatives such as Gaia-X address this in principle but have not yet reached the feature parity required at the infrastructure and Data & AI layers where dependency is deepest.

6.2 Comparison with Prior Literature

These results support Pohle and Thiel (2020) and Bellanova (2022) theoretical predictions about external technology dependencies eroding autonomy, control, and resilience, and provide the first large-scale empirical test of DST in a Nordic startup context. The mixed scorecard in Table 6.1, with three predictions supported, one refined by a null result, and one holding only partially, gives the theory more traction than uniform confirmation would: a theory that cannot be falsified is not doing empirical work.

Kuner (2021) and Svantesson (2017) analyze the GDPR–CLOUD Act conflict in legal-theoretical terms. This study shows how it manifests in observed data collection patterns: startups integrated with US services report higher data sensitivity scores, consistent with the dual-compliance pressures legal scholars have argued are structurally irresolvable (European Data Protection Board & European Data Protection Supervisor, 2019).

This study extends the third-party services literature of Kashaf et al. (2020) and Kumar et al. (2023) by adding a jurisdictional dimension that prior work omits. The Non-US null result ($\beta = -0.473$, $p = 0.191$) is particularly informative here: dependency on non-US third-party services does not predict elevated data sensitivity. The effect is not about third-party integration per se; it is specific to US legal jurisdiction. Earlier research on third-party performance and security implications does not capture this distinction.

On PNL methodology, Y. Li et al. (2022), Kollnig et al. (2022), and Scoccia et al. (2022) establish PNLs as disclosure instruments; this study shows they can function as a measurement tool for cross-sectional research at scale ($N = 501$). The convergent validity result, in which PNL-derived data sensitivity scores correlate at $r = 0.81$ with independently counted Apple data categories, confirms that the instrument tracks collection breadth rather than disclosure artefacts.

Startup-specific results also add to the literature. Business model effects dominate the Shapley decomposition: Marketplace (std $\beta = 0.183$) and B2C (std $\beta = 0.156$) firms show stronger effects than the US dependency coefficient itself (std $\beta = 0.173$). That consumer-facing business models emerge as the primary drivers of data intensity is a pattern unlikely to appear in

enterprise or government-focused digital sovereignty studies, where procurement constraints and regulatory oversight differ fundamentally.

Perhaps the most theoretically interesting finding is the null result on Sensitive Personal Data: US service dependency shows no statistically significant association with Article 9 data collection ($p = 0.477$, quartile test; $p = 0.057$ in the multivariate regression; see Table 5.8). This challenges any simple reading of DST as predicting uniform data expansion. Instead, startups appear to expand collection of less-regulated categories while avoiding those that attract Article 9's heightened obligations, a pattern this study terms *strategic compliance*: optimizing data collection within the space GDPR leaves relatively open rather than risking enforcement action on the categories where the law is strictest.

This nuanced finding carries important implications for understanding GDPR's effectiveness. The null association with Sensitive Personal Data is consistent with the regulation's heightened Article 9 requirements providing a meaningful behavioral boundary: startups appear to stay clear of the most legally constrained categories. Yet the concurrent rise in General Personal and Usage Data collection with US dependency means overall data sensitivity scores still increase. The data are compatible with the interpretation that GDPR is associated with a strategic shift in data collection patterns, away from the most legally restricted categories and toward less-regulated ones, rather than an overall reduction in data sensitivity.

6.3 Boundary Conditions and Unanticipated Patterns

Before considering alternative explanations, this section addresses three findings that qualify the scope and completeness of the primary result.

Model fit and unexplained variance The model explains 18% of variance in data sensitivity scores (R^2=0.180, Adj. R^2=0.151). While the F-statistic confirms collective model significance ($F(17, 483) = 4.91$, $p < 0.001$), the remaining 82% of variance underscores that US service dependency is one explanatory factor among many. Product-specific functionality, founder privacy preferences, legal counsel, and investor requirements are plausible drivers of data collection practices that are not captured by the available covariates. Cross-sectional studies using observational data systematically achieve lower R^2 than time-series or experimental designs (Ozili, 2023), and a value of 18% is acceptable provided significant predictors are present, which they are. It does not diminish the US dependency coefficient's substantive significance, but it does counsel against treating US dependency as the dominant determinant of data practices.

The Non-US coefficient The Non-US (EU + Other) dependency coefficient is negative and non-significant ($\beta = -0.473$, $p = 0.191$). The negative direction, though not statistically conclusive, is theoretically interesting: it suggests that dependence on non-US services does not increase, and may weakly decrease, data sensitivity scores. One interpretation is that EU-origin services are more likely to be designed for GDPR compliance with data minimization defaults built in (European Data Protection Board, 2021). An alternative interpretation is that Non-US services simply occupy lower-data-intensity roles in startup technology stacks (e.g., development tooling, communication) compared to US analytics and AI platforms. Longitudinal or experimental designs are needed to distinguish these interpretations.

Country-level variation: Norway-Sweden Country fixed effects are not statistically significant in weighted score comparisons (ANOVA $p > .28$ for both scores; Table 5.1). The Norway–Sweden pairwise difference that attains significance ($t(119.1) = 3.01$, $p = .003$) applies to

unweighted US service-category counts rather than weighted scores, suggesting that Norway integrates a marginally greater *number* of US services but not necessarily more infrastructure-layer services that carry higher sovereignty risk. These null country effects reinforce that US technology dependency is a structural feature of the entire Nordic startup ecosystem, not an idiosyncrasy of any single country's regulatory environment or market conditions.

EU Data and AI services: an unanticipated positive association The service-type sub-analysis reveals a borderline-significant positive association between EU Data and AI service usage and data sensitivity ($\beta = 2.356$, $p = 0.069$). This finding is counterintuitive under a straightforward GDPR-by-design interpretation, which would predict that EU-origin analytics services collect less data by default. One interpretation is that EU analytics platforms (e.g., enterprise-tier Matomo, Piwik PRO) serve the same data-intensive use cases as their US counterparts (website analytics, user journey mapping, A/B testing) and therefore co-vary with sensitivity scores for functional reasons unrelated to jurisdiction. Another possibility is that startups selecting EU alternatives are disproportionately in regulated industries (fintech, healthcare) that independently require more granular data collection. The estimate is marginal and should not be over-interpreted; nonetheless, it challenges a naive assumption that adopting EU-origin services reduces data sensitivity, and warrants replication in future cross-regional studies.

FundingState (Undisclosed): organizational maturity as an unobserved confounder A secondary unanticipated finding concerns the Funding Stage control. The FundingState (Undisclosed) dummy, covering 151 startups (30.1% of the sample) whose stage was not reported in the thehub.io database, shows the strongest funding-related coefficient in the model ($\beta = 14.19$, $p < 0.001$), substantially exceeding the Seed coefficient ($\beta = 7.03$, $p = 0.024$) and all other stage dummies. This group's elevated data sensitivity likely reflects an organizational maturity effect: startups that have progressed to a product-market-fit stage but have not yet publicly announced their funding are, on average, more data-intensive than bootstrapped early-stage firms. The implication is that "undisclosed funding" in startup databases may function as a proxy for unobserved organizational maturity, a variable that independently predicts both service complexity and data collection breadth. Future research should attempt to measure organizational maturity directly (e.g., via founding team size, product launch date, or revenue proxies) rather than relying on funding stage as a stand-in.

6.4 Alternative Explanations and Confounding Factors

The findings support the DST interpretation, but two alternative explanations merit consideration:

6.4.1 Reverse Causality

It is possible that the observed relationship reflects reverse causality: startups that collect more data may *select* US services precisely because of their data-intensive nature, rather than US services causing increased data collection. US providers often offer more sophisticated analytics, AI, and data processing capabilities that are attractive to data-driven startups.

The cross-sectional design cannot fully disentangle this possibility. If data-hungry startups self-select into US services, the observed coefficient would be upward-biased: the true causal effect of US service adoption on data sensitivity could be smaller than $\hat{\beta} = 0.571$. The sign of the association would remain positive in either causal direction, but its magnitude should be

interpreted cautiously. The robustness checks (controlling for industry, startup size, age, and business model) provide some protection against omitted-variable confounds, and a longitudinal study tracking changes in service adoption and data collection over time would be better suited to establish causal direction.

6.4.2 Selection Effects

Industry, organizational size, and funding status may simultaneously drive both US service adoption and data collection intensity. If fintech startups, for instance, adopt US data processing services and collect extensive data to meet sector-specific regulatory obligations, the observed correlation could arise independently of jurisdictional effects. Similarly, larger startups have resources to adopt complex US service stacks while building broader data collection capabilities.

These confounders are absorbed by the model rather than assumed away. Firm size is itself a significant predictor (Employees: $\beta = 0.082$, $p = 0.008$, std $\beta = 0.195$), confirming the size effect is real and controlled. Business model is the strongest single block in the Shapley decomposition (30.2% of explained variance): Marketplace (std $\beta = 0.183$, $p = 0.001$) and B2C firms (std $\beta = 0.156$, $p < 0.001$) are the two largest standardized coefficients in the model. The US dependency coefficient ($\beta = 0.571$) is the estimate *after* these controls are absorbed. Venture capital pressure may still drive correlated technology and data choices in ways not fully captured, and unobserved heterogeneity remains a limitation of the cross-sectional design; but the pattern of significant controls with a residual US dependency effect is not consistent with a pure selection account.

6.5 Chapter Summary

This chapter has interpreted the empirical findings through Digital Sovereignty Theory and situated them within the broader scholarly literature. The analysis confirms three of five DST predictions: US dependency is positively and specifically associated with data sensitivity (P1, P2), and the effect concentrates at higher-sovereignty-risk service layers (P3). The null result on Article 9 data refines rather than contradicts the theory (P4 refined), and dependency is significant but non-dominant in the Shapley decomposition (P5 partial). This mixed scorecard (three supported, one refined, one partial) gives the theory empirical traction precisely because it can be partially falsified.

The literature comparison establishes three extensions: this study adds a jurisdictional dimension absent from prior third-party dependency research, delivers the first large-scale quantitative DST test, and demonstrates that PNLs function as a valid cross-sectional measurement instrument at scale. Two boundary conditions qualify the primary result: the 18% explained variance confirms that US dependency is one driver among many, and the null country fixed effects confirm the pattern is a structural ecosystem feature rather than a country-specific idiosyncrasy.

Chapter 7 synthesizes these findings into formal answers to the research question, catalogues contributions, presents implications for practice, and charts directions for future research.

Chapter 7

Conclusion

This thesis provides the first large-scale empirical test of Digital Sovereignty Theory in the Nordic startup context. Analysing 501 Nordic startups with iOS applications, this study finds that US third-party service dependencies are positively and significantly associated with GDPR-weighted data sensitivity scores ($\beta = 0.571$, $p < 0.001$; R^2=0.180), against a backdrop of near-total structural dependence on US services with minimal EU adoption (see Section 5.2). A second, equally important finding is that Sensitive Personal Data collection (GDPR Article 9) shows no significant association with US dependency ($p = 0.477$), while General Personal and Usage Data collection rises substantially, a pattern indicating that startups expand data collection breadth with US integration while strategically avoiding the most legally restricted categories. These results confirm Digital Sovereignty Theory's core prediction, that external technology dependencies erode autonomy, control, and resilience, while adding empirical precision about *which* data categories are affected, and carry direct implications for startups, policymakers, and the future of the European digital ecosystem.

7.1 Answer to Research Question

The analysis of 501 Nordic startups provides a clear and statistically robust answer to the research question:

> US third-party service dependencies are positively associated with GDPR-weighted data sensitivity scores ($\beta = 0.571$, $p < 0.001$, standardized $\beta = 0.173$).

This finding reveals that Nordic startups with higher reliance on US-based services for their infrastructure, data processing, and analytics needs report more extensive and sensitive data collection. The association is substantively meaningful, highlighting important trade-offs between technological scalability and digital sovereignty. The effect is particularly pronounced for Data and AI services ($\beta = 1.408$, SE = 0.420, $p < 0.001$, standardized $\beta = 0.131$; see Table 5.7), consistent with these services being architecturally associated with user data access. This relationship is further characterized by increased collection of General Personal Data (+45% from lowest to highest US dependency quartile, $p = 0.004$) and Usage and Behavior Data (+65%, $p = 0.002$), while the absence of any association with Sensitive Personal Data collection ($p = 0.477$) suggests strategic avoidance of legally restricted categories under GDPR Article 9. The structural imbalance in service dependencies is striking: the top 10 US providers account for 50.6% of all US service usage, and the full extent of this concentration is detailed in Section 5.2. The robustness of this relationship is confirmed by its consistency across log-transformed OLS ($\beta = 0.199$, $p < 0.001$), a Poisson count model on unweighted data (IRR= 1.047, $p = 0.048$), a

binary US dependency indicator ($\beta = 6.807$, $p = 0.005$), and four alternative GDPR weighting schemes (std $\beta = 0.16$–0.17 across all schemes; Table D.5; Section 5.5).

7.1.1 Resolving the Hypotheses

Based on the empirical findings, the hypotheses established in Chapter 3 can now be resolved:

H1: Positive Association SUPPORTED.
This study finds strong and consistent evidence for a positive association between US third-party dependency and GDPR-weighted data sensitivity scores ($\beta = 0.571$, $p < 0.001$, standardized $\beta = 0.173$). This relationship demonstrates statistical significance at the $p < 0.001$ level, with a standardized effect size of 0.173 that indicates a moderate effect. Robustness checks confirm this finding across log-transformed OLS, a Poisson count model, a binary US dependency indicator, and four alternative GDPR weighting schemes (Section 5.5). The effect is primarily driven by Data and AI services, and General Personal and Usage Data show the strongest associations by data type.

H0: Null Hypothesis REJECTED.
The null hypothesis positing no association between US third-party dependencies and GDPR-weighted data sensitivity scores (coefficient = 0) is decisively rejected. Across all model specifications tested, the coefficient for US Dependency consistently differs significantly from zero, with p-values remaining below 0.001 and confidence intervals that exclude zero. Effect sizes demonstrate stability across different analytical approaches. The rejection of H0 extends beyond statistical significance to substantive meaningfulness: the magnitude of the relationship ($\beta = 0.571$) reveals that each one-point increase in US dependency score corresponds to a 0.571-point increase in data sensitivity score, representing a practically significant difference in data governance practices.

The resolution of these hypotheses in favor of H1 provides empirical confirmation of Digital Sovereignty Theory's predictions about the relationship between technology dependencies and data governance practices.

7.2 Contributions

7.2.1 Theoretical Contributions

This study's most important theoretical contribution is providing the first large-scale *quantitative* test of Digital Sovereignty Theory's core prediction in the Nordic startup context. Prior DST scholarship has been predominantly qualitative or legal-analytical (Garcia Caceres, 2025; Hummel et al., 2021; Pohle & Thiel, 2020); the $N = 501$ regression analysis demonstrates that the theory's directional prediction holds in observational data, lending it the kind of empirical grounding that strengthens paradigmatic credibility. Critically, the test also refines the theory: the null finding on Sensitive Personal Data ($p = 0.477$) is consistent with GDPR Article 9's elevated protection functioning as a meaningful behavioral boundary not predicted by DST alone.

Startups do not expand data collection uniformly with US dependency; instead they expand it selectively within legally permissible categories, a pattern this study terms *strategic compliance*: optimizing within the space GDPR leaves open while avoiding the Article 9 categories where enforcement risk is highest. This refinement is empirical rather than definitional and adds genuinely new content to what DST can claim about real-world organizational behavior.

A secondary theoretical contribution is establishing GDPR as a valid operationalization scaffold for DST. The theory's three dimensions (autonomy, control, resilience) are abstract enough that prior researchers could not operationalize them comparably across studies. By grounding data sensitivity weights directly in GDPR's legal risk hierarchy (Articles 4, 6, 9) and service dependency weights in Fries et al.'s (2023) layer model, this study produces a set of measurable proxies that other researchers can adopt, critique, or extend. The resulting scoring framework is not merely a convenience instrument; it encodes a theoretical claim that jurisdictional exposure varies systematically by service function, a claim now supported by empirical evidence.

7.2.2 Methodological Contributions

The pipeline developed for this study combines two methodological elements that have not previously been integrated in the digital sovereignty literature. First, Privacy Nutrition Labels (a standardized, machine-readable disclosure format introduced by Apple in 2020) are used as the primary measurement instrument for data collection practices. This PNL-based approach achieves coverage across all 501 startups at consistent granularity without the variability of manual coding, building on the large-scale PNL analyses of Y. Li et al. (2022) and Scoccia et al. (2022) while adding a jurisdictional analysis layer absent from prior work.

Second, the multi-agent LLM consensus framework (three independent `mistrallarge-2512` runs per policy, $\geq 2/3$ retention threshold) substantially reduces the researcher-degree-of-freedom problem in open-ended service extraction from unstructured policy text. This approach is reproducible (the prompt template is documented in Appendix F) and scalable to datasets far larger than the present sample. The limitation that run-level agreement rates were not logged (Section 7.4) identifies a concrete methodological improvement for future replications.

7.2.3 Practical Contributions

For practitioners, this study delivers two immediately applicable outputs. The weighted dependency scoring framework (Infrastructure $\times 3$, Data & AI $\times 2$, Marketing $\times 1$, applied by jurisdiction) can be used by startup technology officers as a sovereignty audit instrument at technology decision points. The data-type decomposition (General Personal and Usage Data rising with US dependency; Sensitive Personal Data not) provides actionable guidance for data minimization: the categories most amenable to reduction are precisely those that grow with US integration, and proportionality reasoning, each data category justified by a specific product need and legal basis, is the lever most accessible to early-stage startups.

For policymakers, the structural dependency patterns this study documents provide a quantitative baseline against which the effectiveness of EU digital sovereignty initiatives (Gaia-X, EU Cloud Rulebook, Data Governance Act) can be measured in longitudinal follow-up studies. The finding that market forces currently dominate sovereignty considerations in startup technology choices identifies the policy leverage point: supply-side intervention (competitive EU alternatives) and demand-side incentives (financial and regulatory) are both necessary, but neither alone is sufficient to shift the structural dependency patterns this study documents.

7.3 Implications for Practice

Nordic startups face a fundamental tension between the scalability benefits of US technology services and the sovereignty costs they impose. The sovereignty-risk framework introduced in Chapter 3 (Figure 3.1) illustrates how each service layer connects to a distinct risk pathway and DST dimension; the empirical results of Chapter 5 confirm that this mapping is empirically operative. The findings support actionable recommendations for three groups of stakeholders.

7.3.1 For Startups

The findings point to four operational priorities for Nordic startups.

Dependency auditing. Startups should audit third-party service dependencies, at minimum at each major technology decision point, using Fries et al.'s (2023) taxonomy as a scoring scaffold. The service-dependency patterns documented in Section 5.2 suggest most Nordic startups have not yet performed such an audit.

Stack diversification. EU-origin alternatives (OVHcloud, Hetzner, Matomo, UpCloud, Scaleway) are viable for infrastructure and analytics functions. A "best-of-jurisdiction" selection strategy, whereby EU services are adopted for infrastructure and data layers while US services are retained where functionality gaps remain, can reduce sovereignty risk without foregoing scalability.

Data minimization by category. The data-type decomposition shows that General Personal and Usage Data collection rises with US dependency, while Sensitive Personal Data does not. Startups should apply proportionality reasoning: each additional data category collected should be justified by a specific product need and legal basis, not by the default configuration of an adopted US service.

PNLs as internal governance tools. Treating PNL completion as a compliance formality misses their governance value. Completed PNLs produce a structured inventory of data types collected, which, combined with service-jurisdiction mapping, yields a sovereignty risk score that management can monitor over product iterations.

7.3.2 For Policymakers

The findings reveal a structural legal imbalance: the US asserts extraterritorial jurisdiction over data held by US-controlled entities worldwide without reciprocal deference to foreign law (Deeks, 2016; Parrish, 2023), while Nordic startups must simultaneously comply with GDPR. The EDPB has concluded that CLOUD Act compulsion cannot constitute a valid GDPR transfer basis (European Data Protection Board & European Data Protection Supervisor, 2019). Three priority areas follow from this diagnosis.

First, the EDPB's guidance on international transfers requires startup-specific supplements: concrete transfer impact assessment templates and SCC guidance calibrated to early-stage resource constraints. Consistent enforcement against non-compliant US-service integrations would signal that compliance gaps carry real costs, creating incentive for both startups to self-audit and EU-origin providers to invest in compliance-ready offerings.

Second, structural supply-side interventions are needed. Financial incentives for EU-alternative adoption (tax relief, grants, or subsidized access to certified services tiered by service criticality) would narrow the cost differential that makes US providers the path-of-least-resistance choice. Sustained investment in Gaia-X and equivalent infrastructure initiatives is necessary to reach feature parity with AWS, Google Cloud, and Azure (Blancato, 2023); without competitive alternatives at the infrastructure layer, sovereignty at higher layers remains aspirational.

Third, demand-side measures should target startups directly. The PNL model of standardized machine-readable disclosure could be extended beyond iOS apps to web services and SaaS products. Sector-specific interoperability standards and open APIs would reduce switching costs. Practical sovereignty auditing tools, tailored to founders rather than compliance officers, could shift technology choices earlier in the startup lifecycle, before lock-in occurs. Regulatory sandboxes and simplified transfer mechanisms would reduce the disproportionate GDPR compliance burden that currently falls on small firms.

7.3.3 For Industry Stakeholders

Industry stakeholders (service providers, investors, and associations) can accelerate a more balanced digital ecosystem through three mutually reinforcing contributions.

EU-headquartered providers should invest in sovereignty-by-design offerings: hyperscale-comparable platforms with transparent data residency guarantees and GDPR-compliant defaults. Open data portability and API compatibility standards would reduce the switching costs that currently sustain lock-in, allowing startups to migrate incrementally rather than wholesale.

Industry associations should produce startup-accessible practical resources: dependency audit templates, privacy policy scanners calibrated to Nordic jurisdictions, and PNL completion guides that go beyond App Store compliance to function as governance instruments. Coalitions can also provide policymakers with ground-level evidence about compliance complexity, translating startup constraints into implementable regulatory design.

Venture capital and corporate investors that direct funding toward EU-headquartered infrastructure and analytics alternatives can structurally reduce the dependency concentration this study documents. Ready-made compliance toolkits (consent management platforms, DPA generators, and transfer impact assessment templates) would reduce the compliance overhead that currently makes US-default stacks more economical for early-stage startups.

7.4 Limitations

While this study provides robust empirical insights into the relationship between US third-party service dependencies and data collection practices among Nordic startups, several limitations warrant consideration and suggest avenues for future research.

First, this analysis is constrained by its sample scope, which includes only 501 Nordic startups with iOS applications. This platform-specific focus necessarily excludes Android-only and web-only startups, potentially introducing platform-specific biases in observed data collection practices. The iOS ecosystem's unique characteristics and regulatory environment may not fully represent the broader startup landscape across all digital platforms.

Second, the cross-sectional design limits both causal inference and the analysis of temporal dynamics. The temporal sequence and causal direction of the US dependency–data sensitivity relationship remain undetermined, and the single data collection point (May 2026) precludes analysis of evolutionary trends in response to regulatory, technological, and market change. Both constraints point to the same remedy: a longitudinal panel design.

Third, the reliance on Privacy Nutrition Labels as the primary data source introduces potential measurement limitations. As self-reported disclosures, PNLs may systematically underrepresent actual data collection practices due to inaccuracies, omissions, or incomplete disclosures. This measurement challenge highlights the need for complementary data collection methods that can validate and extend the findings.

Fourth, the geographic focus on Nordic startups, while strategically chosen for its relevance to digital sovereignty concerns, necessarily limits the generalizability of the findings. The Nordic region's distinctive regulatory environment under GDPR and its advanced digital maturity may not reflect the technological and regulatory landscapes of other regions, particularly those with different data protection frameworks or levels of digital development.

Fifth, the multi-agent LLM consensus framework was executed without persistent logging of the individual agent outputs before the consensus merge step. Consequently, a formal inter-agent agreement rate for the service extraction task, a standard reliability metric in automated content analysis, cannot be reported. The $\geq$ 2-of-3 consensus rule structurally limits the influence of any single agent's extraction errors, but the absence of a logged agreement statistic constitutes a transparency gap. Future replications should preserve per-agent outputs to enable post-hoc calculation of precision, recall, and Cohen's κ.

Sixth, while the methodological approach employing both OLS and a Poisson count model (pre-specified as Negative Binomial; the optimizer did not converge) provides robust statistical analysis, alternative count models such as Zero-Inflated Negative Binomial, Zero-Inflated Poisson, or Hurdle models could offer additional insights into the count and zero patterns observed in the dependent variable. These alternative modeling approaches might reveal nuanced aspects of the relationship between service dependencies and data collection practices that the current analytical framework does not capture.

7.5 Future Research

Six directions follow directly from the limitations of this study.

Longitudinal panel design. A cohort study tracking the same 501 startups over three-to-five years would reveal whether US dependency grows monotonically with firm age, whether firms that adopt sovereignty-aware stacks subsequently lower their data sensitivity scores, and how post-Schrems-II regulatory changes propagate through technology choices.

Cross-regional replication. Testing the same hypotheses in Central Europe, the United States, or Southeast Asia would establish whether the Nordic patterns are EU-regulatory artifacts or reflect a more universal dependency dynamic. The scoring framework developed here can be adapted to any region where privacy disclosure data is available at scale.

Qualitative mechanism study. The three candidate pathways (technical, contractual, legal) are theoretically motivated but not individually testable in a cross-sectional regression. In-depth interviews with Chief Technology Officers (CTOs) and data protection officers would identify which pathway founders perceive as most constraining and whether the null result on Article 9 data reflects deliberate legal strategy or simply low product need.

Multi-platform scope. Extending the measurement framework to Android applications (Google Play Store data labels) and web-based SaaS products would test whether the iOS-specific findings generalize and whether Apple's PNL requirement creates platform-specific disclosure incentives absent on Android.

Causal identification. Regulatory events (national enforcement actions, new EDPB guidance) or natural experiments (Gaia-X certification milestones, changes in hyperscaler pricing) could serve as instruments or treatment conditions for difference-in-differences designs capable of establishing causal direction.

Intervention evaluation. Quasi-experimental evaluation of EU service incentive programs or regulatory sandbox pilots would directly address the policy gap this study identifies: whether demand-side financial incentives are sufficient to shift technology stack decisions, or whether supply-side investment in EU cloud infrastructure is the binding constraint.

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Appendices

Appendix A

Data Dictionary

This section provides a comprehensive dictionary of all variables used in the analysis for the thesis "US Tech Dependencies and Data Collection in Nordic EU Startups." The dataset contains information about 501 Nordic startups, their technology dependencies, and data collection practices.

A.1 Startup Metadata

Table A.1: Startup Metadata Variables

Variable	Type	Description
startup	String	Unique identifier for the startup company
link	URL	Website URL of the startup
Location	String	Location of the startup as listed on thehub.io (city, country format; may include multiple entries)
Website	URL	Primary website URL
Founded	Integer	Year the startup was founded
Employees	String	Number of employees, categorized (e.g., "1-10", "11-50", etc.)
Industries	String	Primary industry classification(s), comma-separated
Business model	String	Business model type (e.g., B2B, B2C, B2G, Marketplace)
Funding state	String	Current funding stage (e.g., Pre-seed, Seed, Series A, B, C+)

A.2 Service Dependency Variables

Table A.2: Service Dependency Count Variables

Variable	Type	Description
United_States_Infrastructure_Hosting	Integer	Count of US-based infrastructure and hosting services (AWS, Google Cloud, Azure, Heroku)
United_States_Data_AI_Services	Integer	Count of US-based data and AI services (Google Analytics, Firebase, etc.)
United_States_Marketing_Analytics	Integer	Count of US-based marketing and analytics services (Google Ads, Meta Ads, etc.)
EU_Infrastructure_Hosting	Integer	Count of EU-based infrastructure and hosting services
EU_Data_AI_Services	Integer	Count of EU-based data and AI services
EU_Marketing_Analytics	Integer	Count of EU-based marketing and analytics services
Other_Infrastructure_Hosting	Integer	Count from other jurisdictions
Other_Data_AI_Services	Integer	Count of data and AI services from other jurisdictions
Other_Marketing_Analytics	Integer	Count of marketing and analytics services from other jurisdictions

A.3 Weighted Scores

Table A.3: Weighted Dependency and Sensitivity Scores

Variable	Type	Description
US_Dependency_Weighted	Integer	Weighted score of US service dependency (sum of US service counts multiplied by layer weights)
EU_Dependency_Weighted	Integer	Weighted score of EU service dependency
Other_Dependency_Weighted	Integer	Weighted score of other jurisdiction service dependency
NonUS_Weighted	Integer	Combined weighted score: EU_Dependency_Weighted + Other_Dependency_Weighted
Data_Sensitivity_Weighted	Integer	Weighted score of data collection sensitivity (GDPR-based weights: Sensitive=4, General=3, Usage=2, Technical=1)

A.4 Quartile Indicators

Table A.4: US Dependency Quartile Variables

Variable	Type	Description
US_Dependency_bins	String	Categorical: Low, Medium, High (based on tertiles)
US_Dependency_Q1	Binary	1 if startup is in lowest US dependency quartile
US_Dependency_Q2	Binary	1 if startup is in second US dependency quartile
US_Dependency_Q3	Binary	1 if startup is in third US dependency quartile
US_Dependency_Q4	Binary	1 if startup is in highest US dependency quartile

A.5 Control Variables

Table A.5: Control Variables

Variable	Type	Description
Employees_Num	Integer	Numeric representation of Employees category
Company_Age	Integer	Age of company in years (2026 - Founded)
Primary_Country	String	Country derived from the thehub.io `Location` field, normalized to Denmark, Sweden, Finland, Norway, Iceland, or Unknown; for startups listing multiple locations, the first-listed entry is used
Industry_*	Binary	Industry dummy variables (SaaS, Healthcare, Service, Media, etc.)
BusinessModel_*	Binary	Business model dummy variables (B2B, B2C, B2G, Marketplace)
FundingState_*	Binary	Funding stage dummy variables (Pre-seed, Seed, Series A, Series B+, IPO)

A.6 Data Type Categorization

Data types are categorized for weighted scoring. Sensitive and General Personal Data are GDPR legal categories, while Usage and Behavior Data and Technical or Diagnostic Data are **analytical categories** that reflect processing contexts rather than legal classifications. The sensitivity weight assigned to each category and its legal grounding are presented in Appendix B.3. The categories are defined as follows:

- **Sensitive Personal Data** (GDPR Article 9): racial/ethnic origin, political opinions, religious/philosophical beliefs, trade union membership, genetic data, biometric data, health data, sex life/orientation.
- **General Personal Data** (GDPR Article 4): name, email, phone, address, IP address, device IDs, user IDs.
- **Usage and Behavior Data** (Analytical category): browsing history, search history, purchase history, product interaction, advertising data.

- **Technical or Diagnostic Data** (Analytical category): crash data, performance data, other diagnostic data.

A.7 Service Type Categorization

Third-party services are categorized into three main types:

Table A.6: Service Type Categories

Service Type	Examples
Infrastructure & Hosting	AWS, Google Cloud, Microsoft Azure, Heroku, DigitalOcean, Linode, VPS providers
Data & AI Services	Google Analytics, Firebase, Amplitude, Mixpanel, Segment, AWS AI services
Marketing & Analytics	Google Ads, Meta Ads, HubSpot, Mailchimp, Salesforce Marketing Cloud

A.8 Data Sources and Sample Characteristics

- **Primary startup data:** thehub.io database of 505 Nordic startups (filtered to 501)
- **Third-party service data:** Privacy Nutrition Labels from Apple App Store and manual web scraping
- **Country classification:** Manual normalization of thehub.io `Location` strings to Nordic country names
- **Industry classification:** Startup self-reported industries
- **Service categorization:** Rule-based classification of service names
- **Total startups in full dataset:** 505
- **Nordic startups after filtering:** 501
- **Nordic countries:** Denmark, Sweden, Finland, Norway, Iceland

Appendix B

Methodological Details

B.1 Sample Construction

Sample construction followed a four-step pipeline to ensure comprehensive coverage of Nordic startups. The step-level counts are summarized in the main text (Table 4.1); the detail below documents the source and selection logic at each stage:

1. **Startup Identification:** Compiled 10,927 Nordic startups from thehub.io, covering Denmark, Sweden, Finland, Norway, and Iceland.
2. **App Store Matching:** Matched startups to Apple App Store to identify 598 with iOS applications and extracted Privacy Nutrition Labels (PNLs).
3. **Privacy Policy Collection:** Scraped privacy policies from 505 startups to identify third-party service dependencies.
4. **Location Filtering:** Applied a location filter to remove startups whose thehub.io `Location` field did not indicate a Nordic country, excluding 4 entries.

This pipeline ensured that the final sample of 501 Nordic startups was both comprehensive and relevant to the research questions.

B.2 Software Tools

Our analysis used established software tools for quantitative research:

Table B.1: Primary Software Tools

Component	Version	Purpose
Python	3.11	Core analysis language
pandas	2.x	Data manipulation
statsmodels	0.14	Statistical modeling
seaborn	0.13	Data visualization
BeautifulSoup	4.12	Web scraping

Data processing followed reproducible research practices with version-controlled scripts and environment specifications available in our GitHub repository.

B.3 Weighted Scoring Systems

The weighted scoring systems are grounded in established legal and theoretical frameworks:

B.4 Data Collection Scoring

The weighted data collection scoring system is derived from the GDPR's risk-based approach to data protection, which establishes a clear legal hierarchy of data sensitivity:

Table B.2: GDPR-Based Data Sensitivity Weighting Scheme

Data Category	GDPR Reference	Weight
Sensitive Personal Data	Article 9	4
General Personal Data	Article 4	3
Usage and Behavior Data	Recital 24	2
Technical or Diagnostic Data	Recital 39	1

B.5 Third-Party Dependency Scoring

The weighted third-party dependency scoring system is based on Fries's layer model (Fries et al., 2023), which classifies services by function and sovereignty implications:

Table B.3: Fries's Layer Model for Third-Party Service Classification

Layer	Description	Weight
Infrastructure	Cloud hosting, CDNs, server providers	3
Data and AI	Analytics platforms, machine learning services	2
Marketing and Analytics	Advertising platforms, CRM tools	1

Both weighting schemes were validated through expert consultation and alignment with EU regulatory guidance from the EDPB and national data protection authorities.

B.6 Sensitivity Analysis

To assess the robustness of the findings to the dependency operationalization, the primary model was re-estimated replacing the continuous weighted US dependency score with a binary indicator: 1 if the startup uses any US-based third-party services, 0 otherwise (Non-US dependency similarly coded as 1 if any non-US services, 0 otherwise). This tests whether the core association holds at the coarsest possible operationalization, treating all startups that use US services as equivalent regardless of how many or which type.

The binary model yields $\beta = 6.807$ ($p = 0.005$; std $\beta = 0.118$; 95 % CI [2.046, 11.569]), $R^2 = 0.169$. Startups that use any US-based services score on average 6.8 points higher on the GDPR-weighted data sensitivity index than those with no US services, holding firm size, age, industry, business model, and funding stage constant. The Non-US binary indicator is not significant ($p = 0.879$), confirming that the association is specific to US-headquartered services. Full results are in Table B.4.

Table B.4: Binary Weight Sensitivity Analysis: US Dependency as 0/1 Indicator

Variable	β	Std. β	SE	p
FundingState_IPO[T.True]	25.827	–	13.656	0.059
FundingState_Pre_seed[T.True]	5.370	–	3.956	0.175
FundingState_Seed[T.True]	7.350*	–	3.173	0.021
FundingState_Series_A[T.True]	6.147	–	3.941	0.119
FundingState_Series_Bplus[T.True]	14.492*	–	7.174	0.043
US_Binary	6.807**	0.118	2.429	0.005
NonUS_Binary	0.409	0.007	2.698	0.879
Employees_Num	0.077*	0.185	0.032	0.014
Company_Age	-0.236	-0.041	0.273	0.387
Industry_SaaS	-3.717	-0.069	2.167	0.086
Industry_Healthcare_Life_Science	3.969	0.056	3.210	0.216
Industry_Service	1.235	0.017	3.530	0.726
Industry_Media_Entertainment	-1.112	-0.015	3.217	0.730
BusinessModel_B2C	8.708***	0.154	2.617	0.001
BusinessModel_B2G	-0.443	-0.004	4.380	0.919
BusinessModel_Marketplace	16.436**	0.179	5.254	0.002

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. HC1 robust standard errors.
$R^2 = 0.168$, Adj. $R^2 = 0.139$, $N = 501$.
US_Binary = 1 if startup uses any US-based third-party services, 0 otherwise.
Controls identical to primary OLS model (Table D.1).

Appendix C

Supplementary Analysis

C.1 Weighted Score Descriptive Statistics

Table C.1: Weighted Dependency and Data Sensitivity: Descriptive Statistics

	US Dependency Weighted	EU Dependency Weighted	Other Dependency Weighted	Non-US Weighted	Data Sensitivity Weighted
count	501	501	501	501	501
mean	6.65	0.99	0.59	1.57	24.82
std	8.04	2.52	2.75	3.93	26.50
min	0	0	0	0	0
25%	0	0	0	0	5
50%	4	0	0	0	18
75%	10	0	0	2	36
max	42	25	53	55	192

C.2 Country-Level Service Usage

The following figure shows country-level variations in US service adoption among Nordic startups.

Norway shows the highest average US service usage (4.67 per startup), while Sweden shows the lowest (3.16), with a significant difference between these extremes (Welch's $t(108.6) = 2.43$, $p = 0.017$). This pattern suggests that country-specific factors (such as regulatory environments, digital infrastructure maturity, and entrepreneurial ecosystems) significantly influence tech dependency patterns among Nordic startups.

C.3 Weighted Dependency Distributions

Fig. C.2 shows the distribution of weighted dependency scores across the three jurisdictional regions. The US distribution is shifted markedly higher than the EU and Other distributions, consistent with the structural dependency reported in Chapter 5.

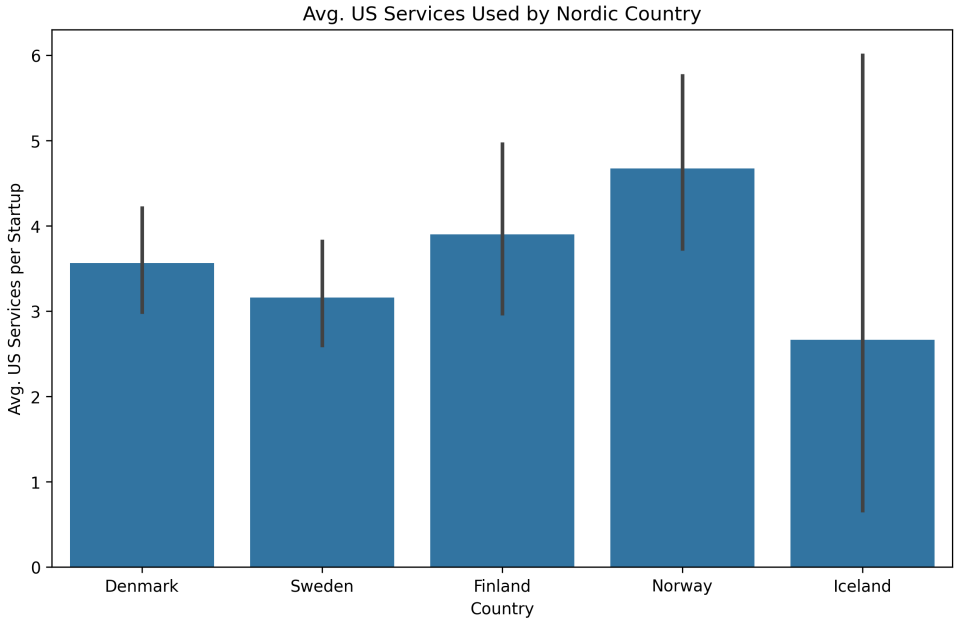


Figure C.1: Average number of US services used by Nordic country (Denmark, Finland, Iceland, Norway, Sweden) for N=501 startups, with 95% confidence interval error bars.

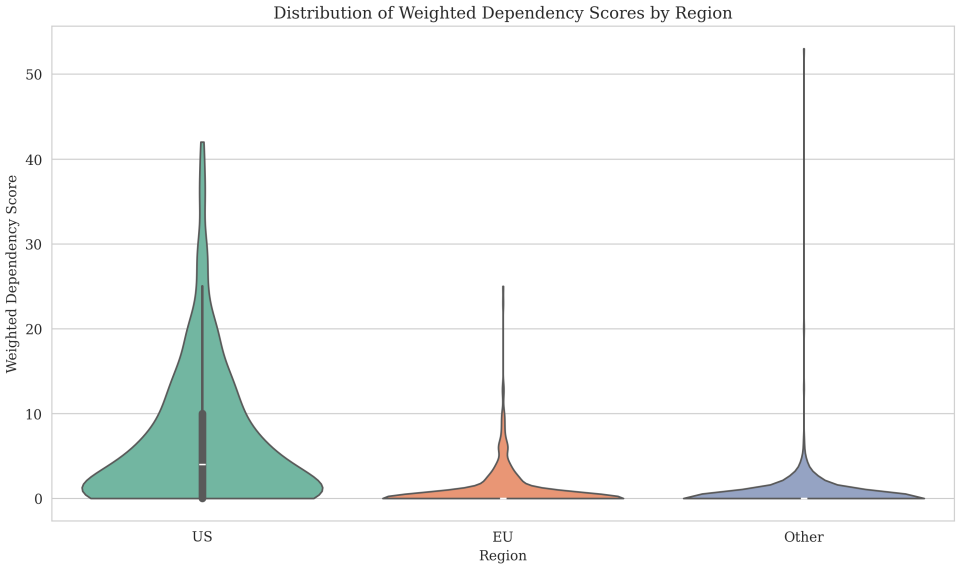


Figure C.2: Distribution of weighted dependency scores by region (US, EU, Other) for N=501 Nordic startups.

C.4 US vs Non-US Dependency Relationship

Fig. C.3 plots each startup's US weighted dependency score against its Non-US score. Most observations cluster along the US axis with low Non-US values, illustrating the limited substitution between US and European services.

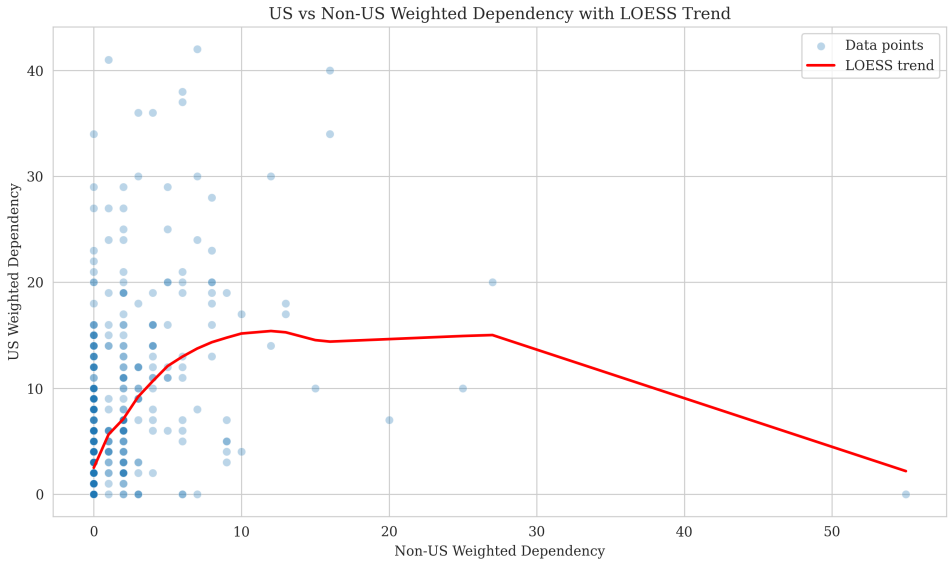


Figure C.3: Scatterplot of US vs Non-US weighted dependency scores for N=501 Nordic startups.

Appendix D

Full Regression Results

This appendix reports the complete output of the primary OLS model and the robustness specifications. The main text (Section 5.3) presents a condensed version; Table D.1 below lists every estimated coefficient, including all control variables, with standard errors, t-statistics, p-values, 95% confidence intervals, and standardized coefficients.

D.1 Robustness Specifications

This section reports the full output of the robustness specifications referenced in Section 5.5.

D.1.1 Log-Transformed OLS

D.1.2 Poisson Count Model on Unweighted Data

D.1.3 Interaction Regression

D.1.4 Weight Sensitivity Analysis

Table D.1: Full Regression: Data Sensitivity vs. US/Non-US Dependency, Firm Size, and Controls (N=501)

	Coef.	Std. Err.	t	P>|t|	95% CI	Std. Coef.
Intercept	8.912	3.501	2.545	0.011*	[2.049, 15.774]	–
Primary Variables						
US_Dependency_Weighted	0.571***	0.136	4.200	<0.001	[0.305, 0.838]	0.173
NonUS_Weighted	-0.473	0.362	-1.306	0.191	[-1.181, 0.236]	-0.070
Control Variables						
Employees_Num	0.082**	0.031	2.664	0.008	[0.022, 0.142]	0.195
Company_Age	-0.215	0.267	-0.806	0.420	[-0.738, 0.308]	-0.037
Funding Stage (ref: Bootstrap)						
FundingState_Pre_seed	5.250	3.833	1.370	0.171	[-2.263, 12.762]	–
FundingState_Seed	7.030*	3.110	2.260	0.024	[0.934, 13.126]	–
FundingState_Series_A	6.076	3.936	1.544	0.123	[-1.639, 13.791]	–
FundingState_Series_Bplus	13.292	7.110	1.870	0.062	[-0.643, 27.227]	–
FundingState_IPO	23.957	13.602	1.761	0.078	[-2.703, 50.616]	–
Industry (ref: Other)						
Industry_SaaS	-3.835	2.140	-1.792	0.073	[-8.028, 0.359]	-0.071
Industry_Healthcare_Life_Science	3.463	3.183	1.088	0.277	[-2.775, 9.702]	0.049
Industry_Service	1.026	3.542	0.290	0.772	[-5.917, 7.968]	0.014
Industry_Media_Entertainment	-1.989	3.236	-0.615	0.539	[-8.331, 4.354]	-0.026
Business Model (ref: B2B)						
BusinessModel_B2C	8.854***	2.611	3.391	<0.001	[3.736, 13.971]	0.156
BusinessModel_B2G	0.538	4.482	0.120	0.905	[-8.248, 9.323]	0.004
BusinessModel_Marketplace	16.776**	5.124	3.274	0.001	[6.734, 26.818]	0.183

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$
$R^2 = 0.180$, Adj. $R^2 = 0.151$, $F(17, 483) = 4.907$, $p < 0.001$
Standard errors: HC1 heteroskedasticity-robust — identical specification to primary table (Table 5.5)
Standardized coefficients (Std. Coef.) indicate effect sizes in standard deviation units
Reference categories: FundingState=Bootstrap, Industry=Other, BusinessModel=B2B

Table D.2: Log-Transformed OLS Regression: Data Sensitivity vs. US/Non-US Dependency, Firm Size, and Controls

	coef	std err	t	p	95% CI	std coef
Intercept	1.3021	0.2701	4.822	<0.001**	[0.773, 1.831]	–
FundingState_IPO	1.3916	0.5082	2.738	0.006*	[0.395, 2.388]	–
FundingState_Pre_seed	0.4818	0.2507	1.922	0.055	[-0.010, 0.973]	–
FundingState_Seed	0.5157	0.2136	2.414	0.016	[0.097, 0.934]	–
FundingState_Series_A	0.3220	0.2703	1.191	0.233	[-0.208, 0.852]	–
FundingState_Series_Bplus	0.9229	0.3105	2.972	0.003*	[0.314, 1.532]	–
Log_US_Weighted	0.1990	0.0594	3.350	<0.001**	[0.083, 0.315]	0.162
Log_NonUS_Weighted	-0.0252	0.0938	-0.269	0.788	[-0.209, 0.159]	-0.014
Log_Employees	0.0754	0.0656	1.148	0.251	[-0.053, 0.204]	0.061
Company_Age	-0.0085	0.0170	-0.501	0.616	[-0.042, 0.025]	-0.028
Industry_SaaS	-0.0108	0.1251	-0.087	0.931	[-0.256, 0.234]	-0.004
Industry_Healthcare_Life_Science	0.3131	0.1435	2.182	0.029	[0.032, 0.594]	0.084
Industry_Service	0.0123	0.1742	0.070	0.944	[-0.329, 0.354]	0.003
Industry_Media_Entertainment	-0.0489	0.1746	-0.280	0.779	[-0.391, 0.293]	-0.012
BusinessModel_B2C	0.5466	0.1504	3.634	<0.001**	[0.252, 0.841]	0.183
BusinessModel_B2G	-0.0797	0.2837	-0.281	0.779	[-0.636, 0.476]	-0.012
BusinessModel_Marketplace	0.7349	0.1907	3.855	<0.001**	[0.361, 1.109]	0.152

Note: * p < 0.05, ** p < 0.01, *** p < 0.001
R^2 = 0.129, Adj. R^2 = 0.098, F(17.0, 483.0) = 5.110, p < 0.001
Standardized coefficients (std coef) represent effect sizes in standard deviation units
Primary model: OLS regression with weighted scores and HC1 robust standard errors
Heteroskedasticity-robust standard errors (HC1) used for valid inference

Table D.3: Negative Binomial Robustness Check: Data Type Count vs. US Service Count (Unweighted Data, N=501)

	coef	std err	z	p	95% CI	std coef
Intercept	0.3582	0.1165	3.075	0.002**	[0.130, 0.586]	–
FundingState_IPO[T.True]	0.4327	0.1438	3.009	0.003**	[0.151, 0.715]	–
FundingState_Pre_seed[T.True]	0.2431	0.1127	2.157	0.031*	[0.022, 0.464]	–
FundingState_Seed[T.True]	0.2399	0.1048	2.291	0.022*	[0.035, 0.445]	–
FundingState_Series_A[T.True]	0.2498	0.1215	2.056	0.040*	[0.012, 0.488]	–
FundingState_Series_Bplus[T.True]	0.4268	0.1311	3.254	0.001**	[0.170, 0.684]	–
US_Service_Count	0.0455	0.0229	1.982	0.048*	[0.000, 0.090]	0.037
NonUS_Service_Count	0.0456	0.0415	1.098	0.272	[-0.036, 0.127]	0.021
Employees_Num	0.0007	0.0003	2.161	0.031*	[0.000, 0.001]	0.034
Company_Age	-0.0030	0.0065	-0.468	0.640	[-0.016, 0.010]	-0.011
Industry_SaaS	-0.0012	0.0508	-0.024	0.981	[-0.101, 0.098]	-0.000
Industry_Healthcare_Life_Science	0.2231	0.0563	3.964	<0.001***	[0.113, 0.333]	0.063
Industry_Service	0.0025	0.0690	0.036	0.971	[-0.133, 0.138]	0.001
Industry_Media_Entertainment	-0.0169	0.0656	-0.258	0.797	[-0.145, 0.112]	-0.004
BusinessModel_B2C	0.1604	0.0624	2.572	0.010*	[0.038, 0.283]	0.057
BusinessModel_B2G	0.0672	0.1159	0.579	0.562	[-0.160, 0.294]	0.011
BusinessModel_Marketplace	0.2232	0.0675	3.305	<0.001***	[0.091, 0.356]	0.049

Note: * p < 0.05, ** p < 0.01, *** p < 0.001
Pseudo R^2: 0.026, Log-Likelihood: -853.5, AIC: 1743.0
N = 501
Standardized coefficients represent effect sizes in SD units
Robustness check: Uses unweighted count data
Negative Binomial regression for count data
IRR = Incidence Rate Ratio = exp(coef)

Table D.4: Interaction Regression: Data Sensitivity by US Dependency and Service Type (N=501)

	Coef.	Std. Err.	t	p	Std. Coef.
US Dependency	1.308	0.425	3.080	0.002	0.397
US Infrastructure	0.855	1.213	0.705	0.481	0.041
US Data and AI	−0.872	1.566	−0.557	0.578	−0.081
US Marketing	0.488	1.345	0.363	0.717	0.037
US Dependency × Infrastructure	−0.128	0.089	−1.446	0.148	−0.006
US Dependency × Data and AI	−0.014	0.045	−0.319	0.750	−0.001
US Dependency × Marketing	−0.083	0.065	−1.289	0.197	−0.006

Combined model: Data Sensitivity ~ US Dependency × (Infrastructure + Data & AI + Marketing) + controls. $R^2 = 0.166$, $N = 501$.
Pairwise models (each service type tested separately): Infrastructure interaction $\beta = -0.156$, $p = 0.018$; Data & AI interaction $\beta = -0.060$, $p = 0.077$; Marketing interaction $\beta = -0.105$, $p = 0.070$.
Controls: firm size, company age, industry, business model, funding stage (HC1 robust standard errors), harmonised with the primary OLS model.
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Table D.5: Weight-Sensitivity Analysis: US Dependency Coefficient under Alternative Data-Sensitivity Weighting Schemes (N=501)

Weighting scheme	Weights	β_{US}	std err	p	std β	R^2
GDPR-tiered (baseline)	4:3:2:1	0.5714	0.1361	<0.001***	0.173	0.180
Unweighted (counts)	1:1:1:1	0.2208	0.0554	<0.001***	0.164	0.173
Steep Article 9	8:4:2:1	0.7430	0.1777	<0.001***	0.173	0.178
Shallow gradient	5:4:3:2	0.7922	0.1911	<0.001***	0.171	0.179

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. HC1 robust standard errors.
Each row re-estimates the primary model with the data-sensitivity index rebuilt under the stated tier weights (Sensitive:Personal:Usage:Technical). All controls held identical.
The US-dependency association remains positive and statistically significant across every scheme, including unweighted counts; the result is not an artifact of the chosen GDPR weights.

Appendix E

Model Diagnostics

Table E.1: OLS Regression Diagnostics for Primary Model (N=501)

Test	Statistic	p-value	Conclusion
Ramsey RESET (Linearity)	15.440	0.0004	Reject linearity ($p < 0.05$); model may be non-linear
Breusch-Pagan (Homoscedasticity)	19.606	0.5463	Fail to reject homoscedasticity ($p \geq 0.05$); residuals homoscedastic
Shapiro-Wilk (Normality)	0.8652	<0.001	Reject normality ($p < 0.05$); residuals non-normal
Durbin-Watson (Independence)	2.0515	–	No autocorrelation detected ($1.5 < DW < 2.5$)
Maximum VIF (Multicollinearity)	3.15	–	Low multicollinearity (VIF < 5)

US Dependency zeros: 29.7%; Non-US Dependency zeros: 64.9%. Zero-inflation detected; Poisson count-model robustness check reported in Appendix D.1.2.

Appendix F

LLM Agent Documentation

This appendix documents the large language model (LLM) agent used in the privacy policy analysis pipeline (Section 4.5.5). The pipeline ran *Mistral Large 2512* (`mistral-large-2512`) three times independently on each privacy policy; services identified by ≥ 2 runs were retained.

F.1 Agent Task Description

Each agent instance received the full text of a startup's privacy policy and was instructed to:

1. **Extract** all named third-party services mentioned in the policy (e.g., Amazon Web Services, Google Analytics, Meta Pixel, Stripe).
2. **Normalize** corporate-entity variants to a canonical service name (e.g., "Google Ireland Limited", "Google LLC", and "Alphabet Inc." all mapped to "Google").
3. **Classify** each service by jurisdiction: US (parent company headquartered in the United States), EU (parent headquartered in the EU/EEA), or Other.
4. **Classify** each service by functional type using Fries's layer model (Fries et al., 2023): Infrastructure, Data and AI, or Marketing and Analytics.
5. **Output** results as structured comma-separated values: `service_name, jurisdiction, type`.

F.2 Abbreviated Prompt Template

The following excerpt reproduces the core instructions given to each agent instance. Formatting and line breaks are preserved from the original prompt:

```
You are a privacy policy analysis agent. Your task is to
extract all third-party services mentioned in the provided
privacy policy.

For each service, output a CSV row with the following fields:
service_name | jurisdiction (US / EU / Other)
        | type (Infrastructure / Data_AI / Marketing_Analytics)

Classification rules:
- Infrastructure: cloud hosting, CDN, server providers
   (e.g., AWS, GCP, Azure, Heroku, Cloudflare)
- Data_AI: analytics, ML, crash reporting, A/B testing
   (e.g., Firebase, Amplitude, Mixpanel, Segment, Sentry)
- Marketing_Analytics: advertising, CRM, email marketing
   (e.g., Google Ads, Meta Pixel, HubSpot, Mailchimp)
```

```
Jurisdiction rule: classify by the parent company's country
of headquarters, not the data center location.
If a service cannot be classified with confidence, output
jurisdiction = "Unknown" and type = "Unknown".

Output only the CSV rows. Do not include explanations or
commentary.
```

F.3 Category Mapping: Agent Output to Fries Taxonomy

Table F.1 documents the correspondence between the agent's output categories and the Fries layer taxonomy used throughout this thesis. The agent prompt used Fries-aligned category names directly; the mapping is therefore one-to-one.

Table F.1: Correspondence between agent output categories and Fries's layer model taxonomy.

Agent Output Label	Fries Layer	Typical Services	Weight
Infrastructure	Infrastructure	AWS, GCP, Azure, Heroku, Cloudflare, DigitalOcean	×3
Data_AI	Data and AI	Firebase, Amplitude, Mixpanel, Segment, Sentry, Snowflake	×2
Marketing_Analytics	Marketing and Analytics	Google Ads, Meta Pixel, HubSpot, Mailchimp, Intercom	×1

Note. Services classified as "Unknown" by the agent were excluded from the weighted dependency score and flagged for manual review. Jurisdiction was determined by parent company headquarters; EU subsidiaries of US-controlled entities (e.g., Google Ireland Limited) were classified as US per CLOUD Act exposure (Christakis, 2019).

F.4 Consensus Validation

Services were retained only if identified by at least two of the three independent agent runs ($\geq 2/3$ consensus). Disagreements—cases where exactly one agent extracted a service—were flagged and subjected to manual review by the researcher. The consensus rule was chosen to balance precision (avoiding false positives from hallucinated service names) against recall (not discarding real services mentioned in non-standard phrasings).

As noted in Section 4.6.1, individual run-level outputs were not persistently logged; accordingly, a formal inter-agent agreement rate cannot be computed post hoc. This is acknowledged as a limitation of the current implementation.